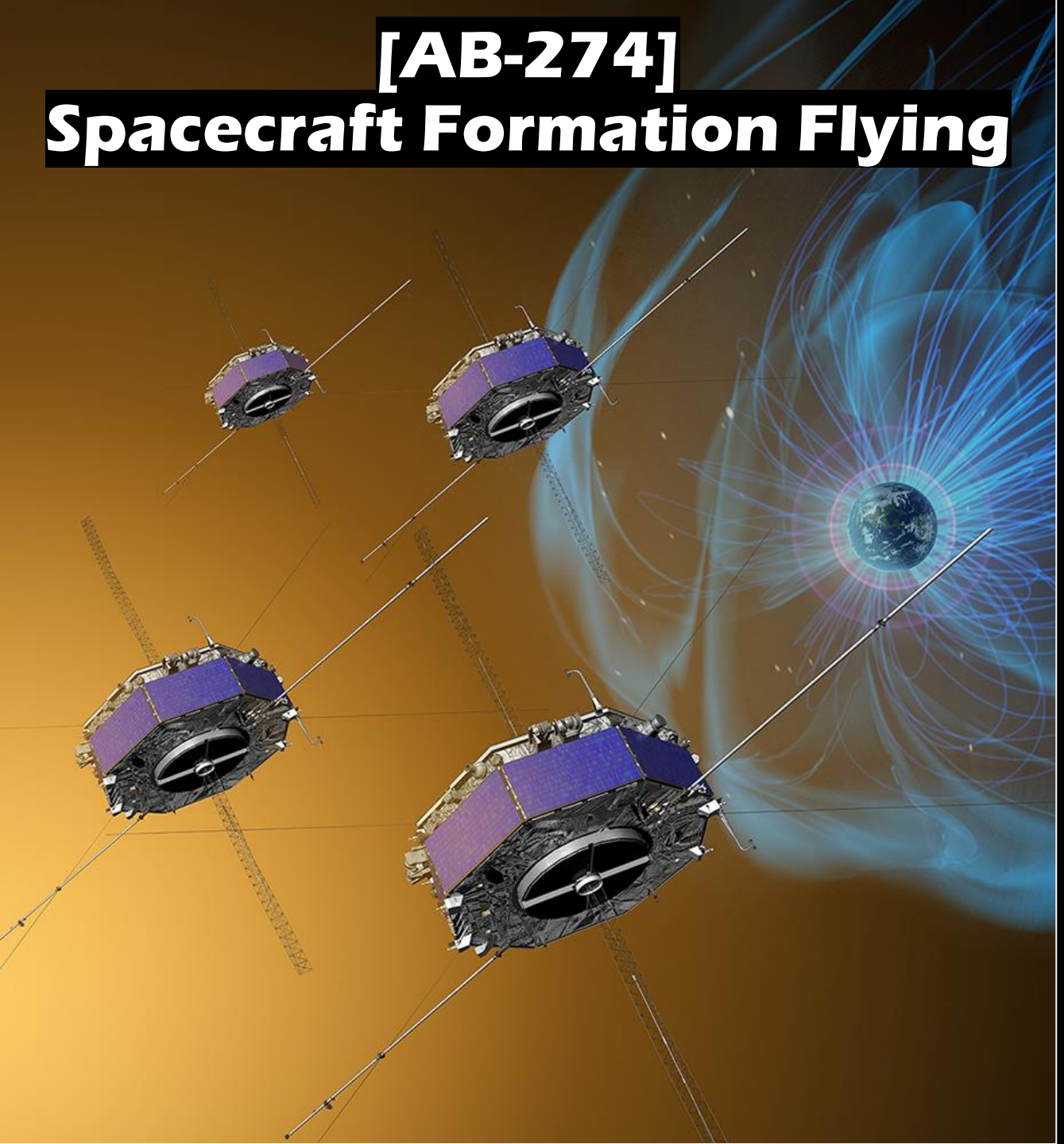


[AB-274] Spacecraft Formation Flying

Lecture #4: "Linear Equations of Relative Dynamics"

Prof. Dr. ***WILLER*** *Gomes dos Santos*

Aerospace Systems Department

willer@ita.br

3rd. Edition, 2022

Class Plan

WEEK	DATE	THEORY
1	09/mar	Introduction, Motivation and Applications.
2	16/mar	Fundamental Astrodynamics.
3	23/mar	Nonlinear Models of Relative Dynamics.
4	**30/mar**	**Linear Equations of Relative Motion.**
5	06/abr	Modeling Relative Motion Using Orbital Elements.
6	13/abr	Modeling Perturbed Relative Motion Using Orbital Elements.
7	20/abr	Perturbation Mitigation (Part I)
8	27/abr	Perturbation Mitigation (Part II)
S	04/mai	**"Little week"**
9	11/mai	Rotation-Translation Coupling
10	18/mai	Fundamentals of Control Systems.
11	25/mai	Formation Control Techniques (Part I)
12	01/jun	Formation Control Techniques (Part II)
13	08/jun	Implementation of delta-V Commands
14	15/jun	Relative Measurements and Navigation (Part I)
15	22/jun	Relative Measurements and Navigation (Part II)
16	29/jun	High-Fidelity Flying Simulation
E	13/jul	Final Examm (Paper presentation)

Previously in Lecture #3…

Nonlinear Models of Relative Dynamics

How can we find a bounded relative motion in the absence of perturbations?

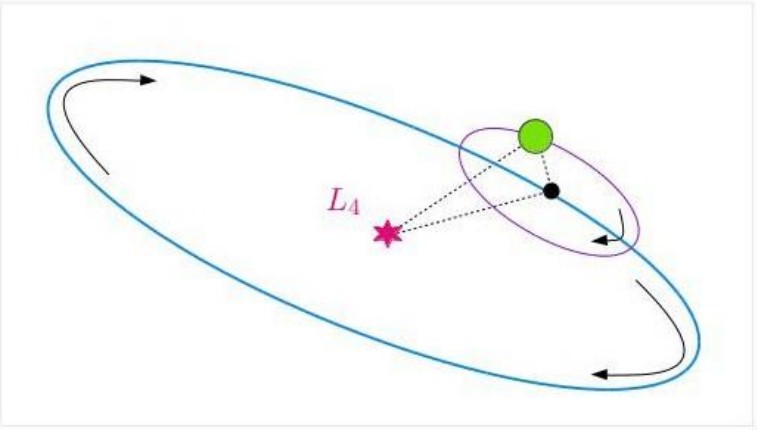
- Through the **Energy Matching Condition** method.

In the presence of initialization errors, how the orbital commensurability can be reestablished?

- Applying a singly thrust impulse (**impulsive formation-keeping**).

What is a libration orbit?

- It is a **periodic motion about the equilibria** when the chief's circular orbit has a commensurable motion.



Summary Lecture #4



- **Introduction**
- The Clohessy–Wiltshire Equations
- Two-impulse Linear Rendezvous
- Accommodating Second-Order Nonlinearities
- Curvilinear vs. Cartesian Relative Coordinates
- Elliptic Reference Orbits
- Review and References



Introduction

- In this class, we present a variety of **linear differential equations** for modeling **relative motion** under the **two-body assumptions**.

- The **systems of equations** for the **description of relative motion** can be **classified** according to the **coordinate system** utilized: **Cartesian or curvilinear**.

- There are also the issues of the choices of the **independent variable** – **time or an angle variable** and the **space** in which the **linearization** is carried out, e.g., physical coordinates vs. a set of orbital elements.

- The focus in this class is on the **physical coordinate description** of relative motion.



Summary Lecture #4



- Introduction
- **The Clohessy–Wiltshire Equations**
- Two-impulse Linear Rendezvous
- Accommodating Second-Order Nonlinearities
- Curvilinear vs. Cartesian Relative Coordinates
- Elliptic Reference Orbits
- Review and References



The Clohessy-Wiltshire Equations

- In the **previous class**, we developed the **nonlinear equations of relative motion** assuming that the **chief's orbit** is **circular**.

- This **simplification** rendered an **autonomous set of nonlinear differential equations**.

- If the **follower's orbit** in the **inertial space** is only **slightly elliptic** and **slightly inclined** with respect to the **chief's orbit**, the **motion of the follower** will appear **very close** to the chief in a **chief-fixed frame**.

- In this case, Eqs. (3.75)–(3.77) may be **linearized** about the **origin** of the **chief-fixed frame**, $\mathcal{L}$, and the **resulting motion** may be **solved** in **closed-form**.

$$\ddot{x} - 2n_0\dot{y} - n_0^2 x = -\frac{\mu(a_0 + x)}{\left[(a_0 + x)^2 + y^2 + z^2\right]^{\frac{3}{2}}} + \frac{\mu}{a_0^2} \qquad \text{Eq. (3.75)}$$

$$\ddot{y} + 2n_0\dot{x} - n_0^2 y = -\frac{\mu y}{\left[(a_0 + x)^2 + y^2 + z^2\right]^{\frac{3}{2}}} \qquad \text{Eq. (3.76)}$$

$$\ddot{z} = -\frac{\mu z}{\left[(a_0 + x)^2 + y^2 + z^2\right]^{\frac{3}{2}}} \qquad \text{Eq. (3.77)}$$

The Clohessy-Wiltshire Equations

- The **linearized equations of motion** are called the **Clohessy–Wiltshire equations (CW)** or the **Hill–Clohessy–Wiltshire equations (HCW)**.

- These **equations** were **developed** by CW in the early **1960s** to analyze **spacecraft rendezvous** (Clohessy and Wiltshire, 1960).

- There are a **number of ways to develop and solve the CW equations**.

- A **straightforward approach** for developing the **CW equations** is to **expand** the right-hand side of Eqs. (3.75)–(3.77) into a **Taylor series** about the **origin**.

Taylor series

$$f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots, \quad \Rightarrow \quad \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

The Clohessy-Wiltshire Equations

- Taking only the **first-order terms** and denoting $n_0 = \sqrt{\mu/a_0^3}$ we get:

$$-\frac{\mu(a_0+x)}{\left[(a_0+x)^2+y^2+z^2\right]^{\frac{3}{2}}} \approx n_0^2(2x-a_0) \qquad \text{Eq. (4.1)}$$

$$-\frac{\mu y}{\left[(a_0+x)^2+y^2+z^2\right]^{\frac{3}{2}}} \approx -n_0^2 y \qquad \text{Eq. (4.2)}$$

$$-\frac{\mu z}{\left[(a_0+x)^2+y^2+z^2\right]^{\frac{3}{2}}} \approx -n_0^2 z \qquad \text{Eq. (4.3)}$$

- **Rearranging and omitting the subscript 0** (so that $n \equiv n_0$ and $a \equiv a_0$) yields

- Equations (4.4)–(4.6) are the **CW equations with no disturbing and/or control accelerations.**

$$\ddot{x} - 2n\dot{y} - 3n^2x = 0 \qquad \text{Eq. (4.4)}$$

$$\ddot{y} + 2n\dot{x} = 0 \qquad \text{Eq. (4.5)}$$

$$\ddot{z} + n^2 z = 0 \qquad \text{Eq. (4.6)}$$

The Clohessy-Wiltshire Equations

- The **nonhomogeneous forms** of Eqs. (4.4)–(4.6) are

$$\ddot{x} - 2n\dot{y} - 3n^2 x = d_x + u_x \qquad \text{Eq. (4.7)}$$

$$\ddot{y} + 2n\dot{x} = d_y + u_y \qquad \text{Eq. (4.8)}$$

$$\ddot{z} + n^2 z = d_z + u_z \qquad \text{Eq. (4.9)}$$

where $[d_x, d_y, d_z]^T$ and $[u_x, u_y, u_z]^T$ are, respectively, the vectors of **environmental perturbations** and **control accelerations**.

The Clohessy-Wiltshire Equations

- One often **normalizes** the **relative coordinates** by the **radius of the reference orbit**, a, and the **angular velocities** by n, so that in **normalized form** **the unforced CW** Equations (4.4)–(4.6) become

$$\bar{x}'' - 2\bar{y}' - 3\bar{x} = 0 \qquad \text{Eq. (4.10)}$$

$$\bar{y}'' + 2\bar{x}' = 0 \qquad \text{Eq. (4.11)}$$

$$\bar{z}'' + \bar{z} = 0 \qquad \text{Eq. (4.12)}$$

where $(\cdot)'$ denotes **differentiation with respect to normalized time**.

The Clohessy-Wiltshire Equations

- It is convenient to write the **LINEAR differential** Equations (4.4)–(4.6) in **state-space form**.
- Choosing the **state vector** $\mathbf{x} = [x, y, z, \dot{x}, \dot{y}, \dot{z}]$, we have

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) \qquad \text{Eq. (4.13)}$$

where A is the **system matrix**, given by

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 3n^2 & 0 & 0 & 0 & 2n & 0 \\ 0 & 0 & 0 & -2n & 0 & 0 \\ 0 & 0 & -n^2 & 0 & 0 & 0 \end{bmatrix}, \qquad \text{Eq. (4.14)}$$

and the **initial conditions** are $\mathbf{x}(t_o) = [x(t_0), y(t_0), z(t_0), \dot{x}(t_0), \dot{y}(t_0), \dot{z}(t_0)]^T$.

- The **eigenvalues** of A are $\{\pm nj, \pm nj, 0, 0\}$, so a **secular mode** is expected to **appear** in the solution.

The Clohessy-Wiltshire Equations

- The **solution of the CW equations** may be **formulated** in terms of the **transition matrix**, $e^{A(t-t_0)}$,

$$\mathbf{x}(t) = \mathrm{e}^{A(t-t_0)}\mathbf{x}(t_0) \qquad \text{Eq. (4.15)}$$

- Following the usual steps for computing the **transition matrix** yields (for $t_0 = 0$),

$$\mathrm{e}^{At} = \begin{bmatrix} 4-3\,\mathrm{c}_{nt} & 0 & 0 & \frac{\mathrm{s}_{nt}}{n} & \frac{2}{n}-\frac{2\mathrm{c}_{nt}}{n} & 0 \\ -6\,nt+6\,\mathrm{s}_{nt} & 1 & 0 & -\frac{2}{n}+\frac{2\mathrm{c}_{nt}}{n} & \frac{4\mathrm{s}_{nt}}{n}-3\,t & 0 \\ 0 & 0 & \mathrm{c}_{nt} & 0 & 0 & \frac{\mathrm{s}_{nt}}{n} \\ 3\,n\mathrm{s}_{nt} & 0 & 0 & \mathrm{c}_{nt} & 2\,\mathrm{s}_{nt} & 0 \\ -6\,n+6\,n\mathrm{c}_{nt} & 0 & 0 & -2\,\mathrm{s}_{nt} & -3+4\,\mathrm{c}_{nt} & 0 \\ 0 & 0 & -n\mathrm{s}_{nt} & 0 & 0 & \mathrm{c}_{nt} \end{bmatrix} \qquad \text{Eq. (4.16)}$$

The Clohessy-Wiltshire Equations

- We can now determine, by **substituting** Eq. (4.16) into Eq. (4.15), the **solutions to the relative position and velocity components**:

$$x(t) = \left[4x(0) + \frac{2\dot{y}(0)}{n}\right] + \frac{\dot{x}(0)}{n}\sin(nt) - \left[3x(0) + \frac{2\dot{y}(0)}{n}\right]\cos(nt) \qquad \text{Eq. (4.17)}$$

$$y(t) = -[6nx(0) + 3\dot{y}(0)]t + \left[y(0) - \frac{2\dot{x}(0)}{n}\right] + \left[6x(0) + \frac{4\dot{y}(0)}{n}\right]\sin(nt) + \frac{2\dot{x}(0)}{n}\cos(nt), \qquad \text{Eq. (4.18)}$$

$$z(t) = \frac{\dot{z}(0)}{n}\sin(nt) + z(0)\cos(nt), \qquad \text{Eq. (4.19)}$$

$$\dot{x}(t) = \dot{x}(0)\cos(nt) + [3x(0)n + 2\dot{y}(0)]\sin(nt) \qquad \text{Eq. (4.20)}$$

$$\dot{y}(t) = -[6nx(0) + 3\dot{y}(0)] + [6x(0)n + 4\dot{y}(0)]\cos(nt) - 2\dot{x}(0)\sin(nt) \qquad \text{Eq. (4.21)}$$

$$\dot{z}(t) = \dot{z}(0)\cos(nt) - z(0)n\sin(nt) \qquad \text{Eq. (4.22)}$$

The Clohessy-Wiltshire Equations

Some observations can be described:

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t)$$

❶ From the dynamical systems perspective, the **linear system** of Eq. (4.13) has **multiple equilibria** $\{x = 0, y = \text{const.}, z = 0\}$, while the **original**, **nonlinear system** has a **continuum of equilibria** shown in Eq. (3.78).

Eq. (3.78)

$$z = 0, (x + a_0)^2 + y^2 = a_0^2$$

There is some **equivalence between the NONLINEAR and LINEAR systems**:

- *In the **linear system**, the **coordinate** y can be written as $y = a\delta\theta$, where $\delta\theta$ is a **curvilinear coordinate** along the circumference of the reference orbit.*

- Hence, $\delta\theta = \text{const.}$ corresponds to an **angular shift** of the **follower along the chief's reference orbit**.

The Clohessy-Wiltshire Equations

❷ Note that the **linearization** has **decoupled** the **out-of-plane, cross-track, motion from the in-plane motion**.

The **cross-track motion** is a **simple harmonic** that can be **cancelled** by selecting the **initial conditions**: $z(0) = \dot{z}(0) = 0$.

❸ The **transverse, along-track, component** shows the presence of **DRIFT**, varying **linearly** with time, implying that the **in-plane motion is unstable**.

Nevertheless, we can find a **stable subspace of the state-space**, turning out in a **stable motion**.

The **initial conditions** which cover the **stable subspace** are given by

$$\dot{y}(0) = -2nx(0)$$

Eq. (4.23)

The Clohessy-Wiltshire Equations

- Choosing **initial conditions** of Eq. (4.23) will yield a **bounded relative motion to first order**, i.e., **only** for the **CW model**.

- It is important to note that the **concept of stability** in the context of the **CW equations** is **local**.

- There exist **initial conditions** which **violate** Eq. (4.23) but **satisfy** the **energy matching condition**, yielding a bounded **1:1 commensurable motion** described by Eqs. (3.75)–(3.77).

$$\ddot{x} - 2n_0\dot{y} - n_0^2 x = -\frac{\mu(a_0 + x)}{\left[(a_0 + x)^2 + y^2 + z^2\right]^{\frac{3}{2}}} + \frac{\mu}{a_0^2} \qquad \text{Eq. (3.75)}$$

$$\ddot{y} + 2n_0\dot{x} - n_0^2 y = -\frac{\mu y}{\left[(a_0 + x)^2 + y^2 + z^2\right]^{\frac{3}{2}}} \qquad \text{Eq. (3.76)}$$

$$\ddot{z} = -\frac{\mu z}{\left[(a_0 + x)^2 + y^2 + z^2\right]^{\frac{3}{2}}} \qquad \text{Eq. (3.77)}$$

- This happens because the **energy matching condition** is a **global boundedness criterion**.

The Clohessy-Wiltshire Equations

- By using the **CW approximation**, we **truncate the physical solution of the problem**, thus introducing an **inherent error**.



- The **local results** **apply only in a small neighborhood of the chief**, and **cannot be taken into account** for drawing conclusions as to properties of the **relative motion** with respect to **circular, two-body orbits**.

The Clohessy-Wiltshire Equations

GEOMETRIC PROPERTIES OF THE CW SOLUTIONS

- If Eq. (4.23) is satisfied, Eqs. (4.17)–(4.19) may be written in the **magnitude-phase form:**

$$\dot{y}(0) = -2nx(0)$$

$$\boxed{\begin{aligned} x(t) &= \rho_x \sin(nt + \alpha_x) \\ y(t) &= \rho_y + 2\rho_x \cos(nt + \alpha_x) \\ z(t) &= \rho_z \sin(nt + \alpha_z) \end{aligned}}$$

Eq. (4.24)

Eq. (4.25)

Eq. (4.26)

where

$$\rho_x = \frac{\sqrt{\dot{x}^2(0) + x^2(0)n^2}}{n}$$

Eq. (4.27)

$$\rho_y = [y(0) - 2\dot{x}(0)/n]$$

Eq. (4.28)

$$\rho_z = \frac{\sqrt{\dot{z}^2(0) + z^2(0)n^2}}{n}$$

Eq. (4.29)

$$\alpha_x = \tan^{-1}\frac{nx(0)}{\dot{x}(0)}$$

Eq. (4.30)

$$\alpha_z = \tan^{-1}\frac{nz(0)}{\dot{z}(0)}$$

Eq. (4.31)

The Clohessy-Wiltshire Equations

- Equations (4.24)–(4.26) constitute a **parametric representation** of an **elliptic cylinder**.
- If

$$\alpha_x = \alpha_z \qquad \text{Eq. (4.32)}$$

then these **equations** become a **parametric representation** of a **3D ellipse**, as shown in Fig. 4.1.

- This **ellipse** is **centered** on the y-axis at $(0, \rho_y, 0)$.
- If

$$\alpha_x = \alpha_z, \quad \rho_z = \sqrt{3}\rho_x \qquad \text{Eq. (4.33)}$$

then the **relative orbit** is a **3D circle with radius** $2\rho_x$.

- Such relative orbit is commonly referred to as the **General Circular Orbit (GCO)**.

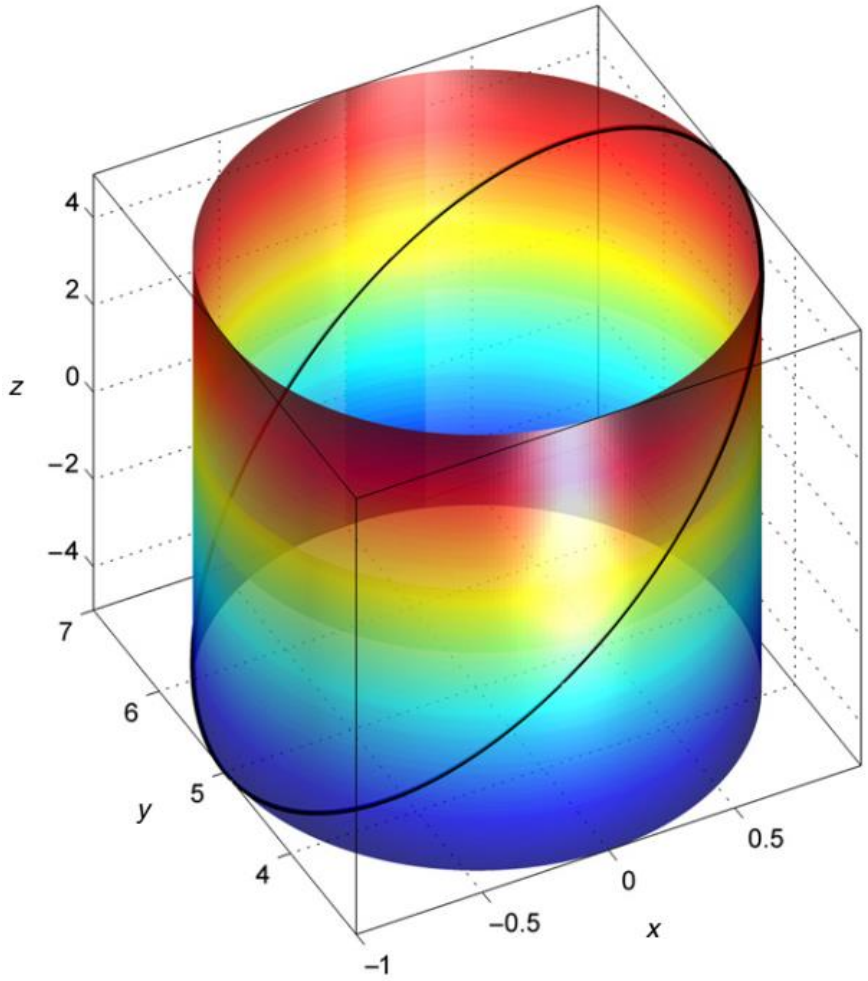


Fig. 4.1 – The bounded relative orbit in the linear approximation lies on a 3D elliptic cylinder (Alfriend et al, 2010).

The Clohessy-Wiltshire Equations

- In general, the xy-**projection of the motion** is a **parametric representation of a 2:1 ellipse** with **semi-major axis** $2\rho_x$, **semi-minor axis** ρ_x and a **constant eccentricity** of $\sqrt{1-\rho_x^2/(4\rho_x^2)} = \sqrt{3}/2 \approx 0{,}87$.

- The **line of apsides** is **rotated** $\pi/2$ radians relative to the y-axis, and the **center of motion** is located at the **offset** $[0, \rho_y]$.

- The xz- and yz-**projections** are, in general, **ovals**.

- There are a **few special cases**, however, where these **projections** become **elliptic**, **circular**, or **linear**.

The Clohessy-Wiltshire Equations

- An **elliptic xz-projection** is obtained if

$$\alpha_z = \pi/2 + \alpha_x \qquad \text{Eq. (4.34)}$$

  and a **circular xz-projection** is obtained if

$$\alpha_z = \pi/2 + \alpha_x, \quad \rho_x = \rho_z \qquad \text{Eq. (4.35)}$$

- A **linear xz-projection** is obtained if Eq. (4.32) holds: $\alpha_x = \alpha_z$ Eq. (4.32)

- Equivalently, an **elliptic yz-projection** is obtained if Eq. (4.32) holds, and a **circular yz-projection** is obtained if

$$\alpha_x = \alpha_z, \quad \rho_z = 2\rho_x \qquad \text{Eq. (4.36)}$$

- The **yz-circular-projection orbit**, referred to as the **Projected Circular Orbit (PCO)** has been quite studied due to its **potential applications**, such as **Earth-imaging**.

- A **linear yz-projection** is obtained if Eq. (4.34) is satisfied.

The Clohessy-Wiltshire Equations

EXAMPLE 4.1

- **Two satellites** are launched aboard a **Delta launcher** and should fly in a **close formation**.
- An **initial velocity impulse** injects the satellites into a **circular LEO of 600 km**.
- An **additional velocity impulse** applied by the follower de-attaches it from the chief so that the **relative position components** are

$$x(0) = 69.78 \text{ km}, \quad y(0) = 139.56 \text{ km}, \quad z(0) = 104.67 \text{ km}$$

and the **relative velocity components** are

$$\dot{x}(0) = 7.5579 \text{ m/s}, \quad \dot{y}(0) = -151.116 \text{ m/s}, \quad \dot{z}(0) = 15.116 \text{ m/s}$$

- **Determine the geometry of the relative orbit.** What **initial conditions** should be **chosen** instead of the given initial conditions to yield **circular xz- and yz-projections**?

The Clohessy-Wiltshire Equations

Solution Of Example 4.1

- The **satellite's orbit** has $a = 600 + 6378 = 6978$ km, $n = \sqrt{3.986 \cdot 10^5 / 6978^3}$ rad/s.

- The **relative position in normalized coordinates** is thus

$$\bar{x}(0) = x(0)/a = 0.01,$$
$$\bar{y}(0) = y(0)/a = 0.02,$$
$$\bar{z}(0) = z(0)/a = 0.015$$

and the **normalized velocity** is

$$\bar{x}'(0) = 0.001;$$
$$\bar{y}' = \cancel{-0.002} - 0.02;$$
$$\bar{z}' = 0.002.$$

- The xy-**projection of the relative orbit** is clearly **elliptic**.

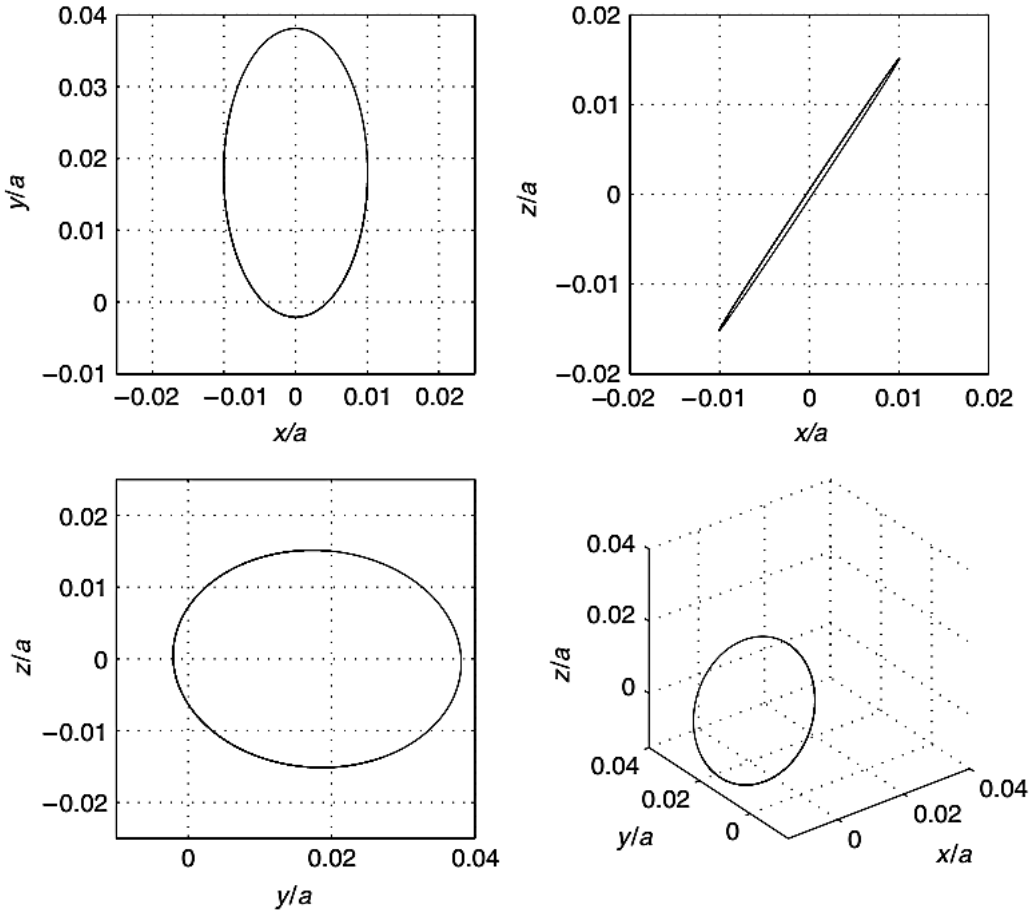


Fig. 4.2 – Geometry of a bounded relative orbit using the CW equations for relative motion (Alfriend et al, 2010).

The Clohessy-Wiltshire Equations

- In order to generate a:
  - **Circular xz-projection** we need $\bar{x}(0) = 0; \bar{x}'(0) = \bar{z}(0) = 0.015; \bar{z}'(0) = 0$, and
  - **Circular yz-projection (PCO)**, we need $\bar{z}(0) = 2\,\bar{x}(0) = 0.02; \bar{x}'(0) = \bar{z}'(0) = 0$.

- The **results** are shown in Fig. 4.3 and Fig. 4.4, respectively.

- As expected, the **circular xz-projection** also yields a **linear yz-projection**.

- Similarly, the **circular yz-projection** yields a **linear xz-projection**.

- In both cases, of course, the **projection on the xy-plane** is a **2:1 ellipse** with **eccentricity** of $\frac{\sqrt{3}}{2} \approx 0.87$.

The Clohessy-Wiltshire Equations

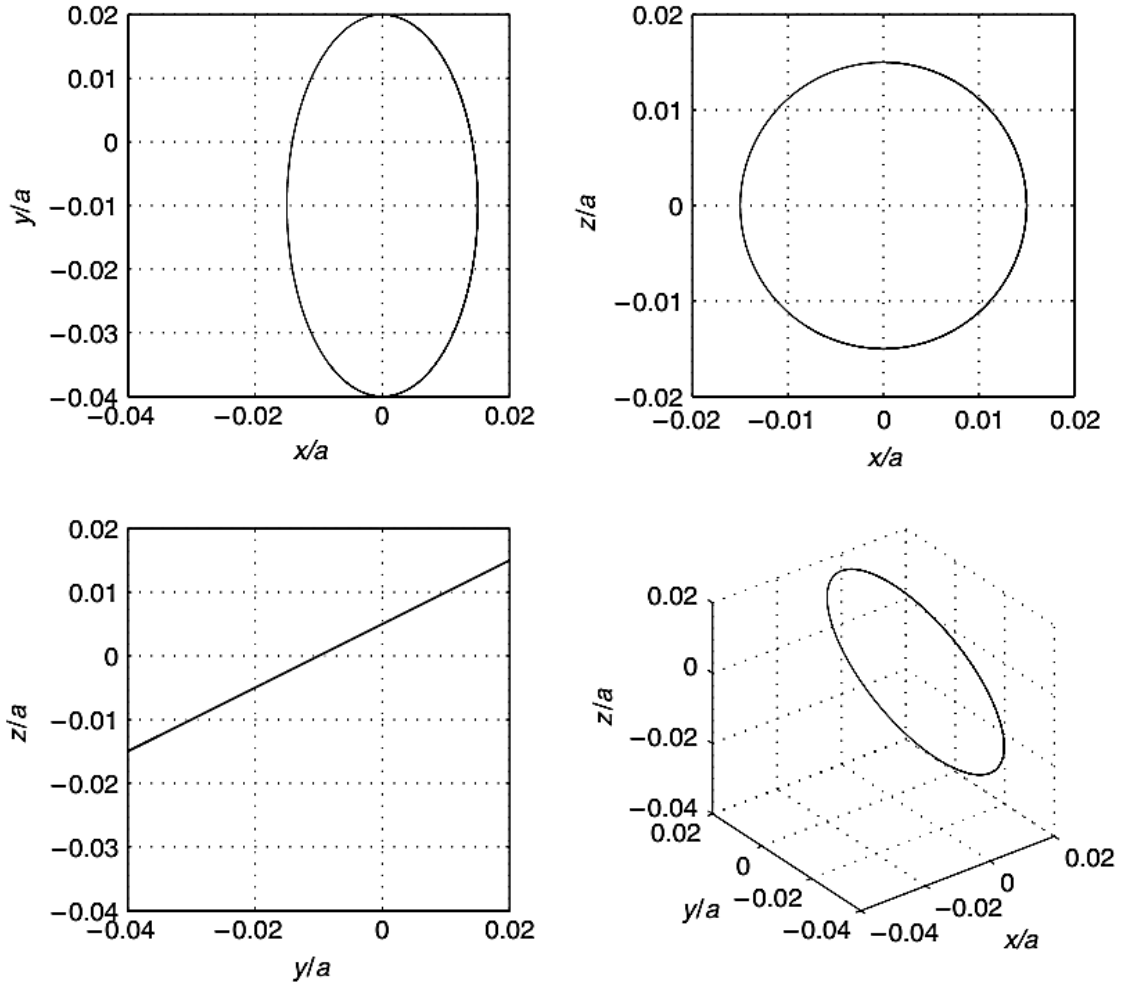


Fig. 4.3 – A circular xz-projection CW relative orbit (Alfriend et al, 2010).

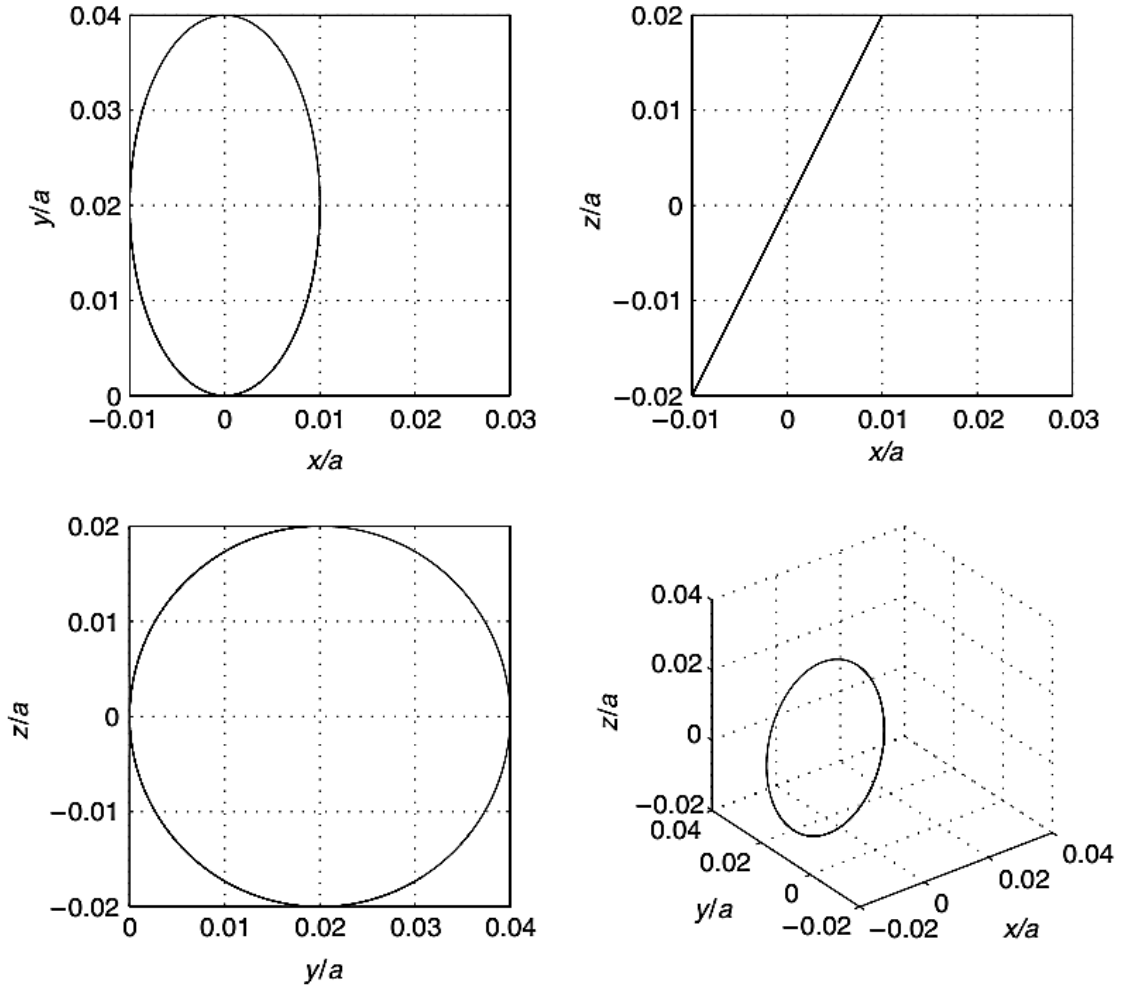


Fig. 4.4 – A circular zy-projection CW relative orbit (Alfriend et al, 2010).

Summary Lecture #4



- Introduction
- The Clohessy–Wiltshire Equations
- **Two-impulse Linear Rendezvous**
- Accommodating Second-Order Nonlinearities
- Curvilinear vs. Cartesian Relative Coordinates
- Elliptic Reference Orbits
- Review and References



Two-Impulse Linear Rendezvous

- The **CW equations are linear**, admitting an **equilibrium** at the **origin**.

- Thus, a **ballistic or thrust-free stable linear relative motion** is possible **if the chief and follower spacecraft coincide**.

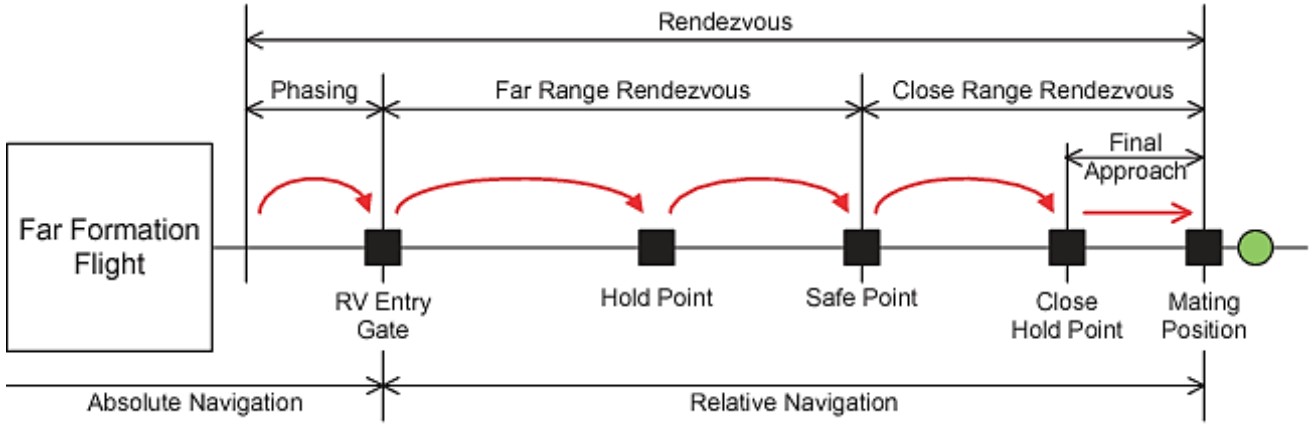
- This is called **RENDEZVOUS**, that is, when **both** the **relative position vector and relative velocity vector** in the **chief-fixed frame** are **anulled**.

- **Rendezvous has many practical applications,** for example, **DOCKING** of the **ATV** (**Automated Transfer Vehicle**) on the **International Space Station** requires a **rendezvous maneuver**.



Two-Impulse Linear Rendezvous

- It is therefore important to **find** **impulsive velocity corrections** that **guarantee** rendezvous.
- This will be **performed** in **two stages**:
  1. Initially find an **impulsive maneuver** that **annuls** the **final relative distance** between the chief and the follower. This **process** is termed **targeting or guidance**.
  2. Determine the **additional impulsive maneuver** required to **annul** the **relative velocity** and **complete** the **rendezvous**.
- This **scheme** is referred to as **two-impulse linear rendezvous**.



Two-Impulse Linear Rendezvous

- Designating the **moment of the targeting maneuver** by $t_0 = 0$, the **velocity impulse**, $\Delta\mathbf{v}_I$, satisfies

$$\Delta\mathbf{v}_I = \mathbf{v}^+(0) - \mathbf{v}^-(0) \qquad \text{Eq. (4.37)}$$

  where $\mathbf{v}^-(0)$ and $\mathbf{v}^+(0)$ are the **relative velocity vectors before and after application of the targeting impulse**, respectively.

- The vector $\mathbf{v}^+(0)$, the **required initial velocity for targeting**, can be found by **substituting** $x(t_f) = y(t_f) = z(t_f) = 0$ into Eqs. (4.17)–(4.19) …

$$x(t) = \left[4x(0) + \frac{2\dot{y}(0)}{n}\right] + \frac{\dot{x}(0)}{n}\sin(nt) - \left[3x(0) + \frac{2\dot{y}(0)}{n}\right]\cos(nt) \qquad \text{Eq. (4.17)}$$

$$y(t) = -[6nx(0) + 3\dot{y}(0)]t + \left[y(0) - \frac{2\dot{x}(0)}{n}\right] + \left[6x(0) + \frac{4\dot{y}(0)}{n}\right]\sin(nt) + \frac{2\dot{x}(0)}{n}\cos(nt), \qquad \text{Eq. (4.18)}$$

$$z(t) = \frac{\dot{z}(0)}{n}\sin(nt) + z(0)\cos(nt), \qquad \text{Eq. (4.19)}$$

Two-Impulse Linear Rendezvous

- … and **solving** for $\dot{x}(0), \dot{y}(0), \dot{z}(0)$, given the **initial relative position components** $x(0), y(0), z(0)$:

$$\mathbf{v}^{+}(0) = \begin{bmatrix} \dot{x}(0) \\ \dot{y}(0) \\ \dot{z}(0) \end{bmatrix}^{+}$$

$$= \begin{bmatrix} -\dfrac{n\left[-4\,x(0)\,s_{nt_f} + 3\,x(0)\,nt_f\,c_{nt_f} + 2\,y(0) - 2\,y(0)\,c_{nt_f}\right]}{-8 + 8\,c_{nt_f} + 3\,nt_f\,s_{nt_f}} \\ -\dfrac{n\left[-14\,x(0) + 14\,x(0)\,c_{nt_f} + 6\,x(0)\,nt_f\,s_{nt_f} - y(0)\,s_{nt_f}\right]}{-8 + 8\,c_{nt_f} + 3\,nt_f\,s_{nt_f}} \\ -z(0)n\cot(nt_f) \end{bmatrix}$$

Eq. (4.38)

$\mathbf{v}^{+}(0)$ **does not exist for all flight times**: $\dot{z}^{+}(0)$ is **singular** for $nt_f = k\pi, k = 0,1, \ldots$, and the **in-plane componentes** $\dot{x}^{+}(0)$ and $\dot{y}^{+}(0)$ are **singular** at $nt_f = 2k\pi, k = 0,1, \ldots$ and at **additional points** satisfying $8c_{nt_f} + 3nt_f s_{nt_f} = 8$ such as $nt_f = 2.8135\pi$ and $nt_f = 4.8906\pi$.

Two-Impulse Linear Rendezvous

- At t_f, when the **follower satellite (CHASER*)** **reaches** the **chief satellite (TARGET*)**, **its relative velocity must be annulled to guarantee rendezvous**.

- Therefore, the **required velocity impulse** at **impact must be equal** in **magnitude** and **opposite** in sign to the **final relative velocity**:

$$\Delta \mathbf{v}_{II} = -\mathbf{v}(t_f) \qquad \text{Eq. (4.39)}$$

where $\mathbf{v}(t_f)$ is the **final relative velocity**, obtained by substituting $t = t_f$ into Eqs. (4.20)–(4.22),

$$\mathbf{v}(t_f) = \begin{bmatrix} \dot{x}(t_f) \\ \dot{y}(t_f) \\ \dot{z}(t_f) \end{bmatrix} = \begin{bmatrix} \dot{x}^+(0)\mathrm{c}_{nt_f} + [3x(0)n + 2\dot{y}^+(0)]\mathrm{s}_{nt_f} \\ -[6nx(0) + 3\dot{y}^+(0)] + [6x(0)n + 4\dot{y}^+(0)]\mathrm{c}_{nt_f} - 2\dot{x}^+(0)\mathrm{s}_{nt_f} \\ \dot{z}^+(0)\mathrm{c}_{nt_f} - z(0)n\mathrm{s}_{nt_f} \end{bmatrix} \qquad \text{Eq. (4.40)}$$

*In Rendezvous literature

Two-Impulse Linear Rendezvous

- The **total velocity change** required for a **two-impulse rendezvous** is

$$\Delta v = \|\Delta \mathbf{v}_I\| + \|\Delta \mathbf{v}_{II}\| \qquad \text{Eq. (4.41)}$$

- In Eq. (4.41), the **vector norm** **depends** on the **type of thrusters used**.

- If the **thrust** is applied using a **single directional thruster**, one must take $\|\cdot\|_2$;

- If a **cluster of body-fixed thrusters** are used, providing the required thrust components in **each** direction, one must substitute $\|\cdot\|_1$ into Eq. (4.41).

Math review

Taxicarb or Manhattan norm

$$\|\boldsymbol{x}\|_1 := \sum_{i=1}^{n} |x_i|$$

Euclidean norm

$$\|\boldsymbol{x}\|_2 := \sqrt{x_1^2 + \cdots + x_n^2}$$

Two-Impulse Linear Rendezvous

EXAMPLE 4.2



- A **satellite** is in a **circular LEO of 600 km**.
- The **Space Shuttle** is launched to **recover** the satellite.
- At $t = 0$, a **laser ranging system** onboard the shuttle **determines** that the **relative separation** between the Shuttle bay and the satellite is

$$x(0) = 69.78 \text{ km}, \quad y(0) = 139.56 \text{ km}, \quad z(0) = 104.67 \text{ km}$$

and that the **relative velocity** is

$$\dot{x}(0) = \dot{y}(0) = \dot{z}(0) = 7.5579 \text{ m/s}$$

- **Determine the components and the magnitude of the impulsive velocity corrections required for rendezvous after 3.22 hours.**

Two-Impulse Linear Rendezvous

Solution of Example 4.2



- Let us first **normalize** the given **quantities**.

- The **satellite's orbit radius** is $a = 600 + 6378 = 6978$ km, and the **mean motion** is $n = \sqrt{3.986 \cdot 10^5/6978^3}$ rad/s.

- The **relative position in normalized coordinates** is therefore

$$x(0)/a = 0.01, y(0)/a = 0.02, z(0)/a = 0.015$$

and the **normalized velocity** is

$$\bar{x}'(0) = \bar{y}'(0) = \bar{z}'(0) = 7.5579/(na) = 0.001.$$

- The **normalized retrieval time** is $3.22 \cdot 3600 \cdot n/(2\pi) = 2$.

Two-Impulse Linear Rendezvous

- Substituting the **data** into Eq. (4.38) gives the **normalized velocity corrections**

$$\mathbf{v}^+(0) = \begin{bmatrix} \dot{x}(0) \\ \dot{y}(0) \\ \dot{z}(0) \end{bmatrix}^+ \qquad \text{Eq. (4.38)}$$

$$= \begin{bmatrix} -\dfrac{n\left[-4\,x(0)\,s_{nt_f} + 3\,x(0)\,nt_f\,c_{nt_f} + 2\,y(0) - 2\,y(0)\,c_{nt_f}\right]}{-8 + 8\,c_{nt_f} + 3\,nt_f\,s_{nt_f}} \\ -\dfrac{n\left[-14\,x(0) + 14\,x(0)\,c_{nt_f} + 6\,x(0)\,nt_f\,s_{nt_f} - y(0)\,s_{nt_f}\right]}{-8 + 8\,c_{nt_f} + 3\,nt_f\,s_{nt_f}} \\ -z(0)n\cot(nt_f) \end{bmatrix}$$

$$\Delta\bar{\mathbf{v}}_I = \begin{bmatrix} -0.00178 \\ -0.01927 \\ 0.005865 \end{bmatrix} \qquad \text{Eq. (4.42)}$$

- The **normalized final velocity change** is computed according to Eq. (4.40),

$$\mathbf{v}(t_f) = \begin{bmatrix} \dot{x}(t_f) \\ \dot{y}(t_f) \\ \dot{z}(t_f) \end{bmatrix} \qquad \text{Eq. (4.40)}$$

$$= \begin{bmatrix} \dot{x}^+(0)c_{nt_f} + [3x(0)n + 2\dot{y}^+(0)]s_{nt_f} \\ -[6nx(0) + 3\dot{y}^+(0)] + [6x(0)n + 4\dot{y}^+(0)]c_{nt_f} - 2\dot{x}^+(0)s_{nt_f} \\ \dot{z}^+(0)c_{nt_f} - z(0)ns_{nt_f} \end{bmatrix}$$

$$\Delta\bar{\mathbf{v}}_{II} = \begin{bmatrix} -0.00562 \\ 0.00173 \\ -0.0165 \end{bmatrix} \qquad \text{Eq. (4.43)}$$

- Assuming a **directional thruster**, the **total normalized velocity change** required is

$$\Delta\bar{v} = \|\Delta\bar{\mathbf{v}}_I\|_2 + \|\Delta\,\bar{\mathbf{v}}_{II}\|_2 = 0.03774.$$

Two-Impulse Linear Rendezvous

- Figure 4.5 depicts the **relative position components**, showing **convergence to zero relative position** at the **normalized time** $\bar{t} = 2$, as expected.

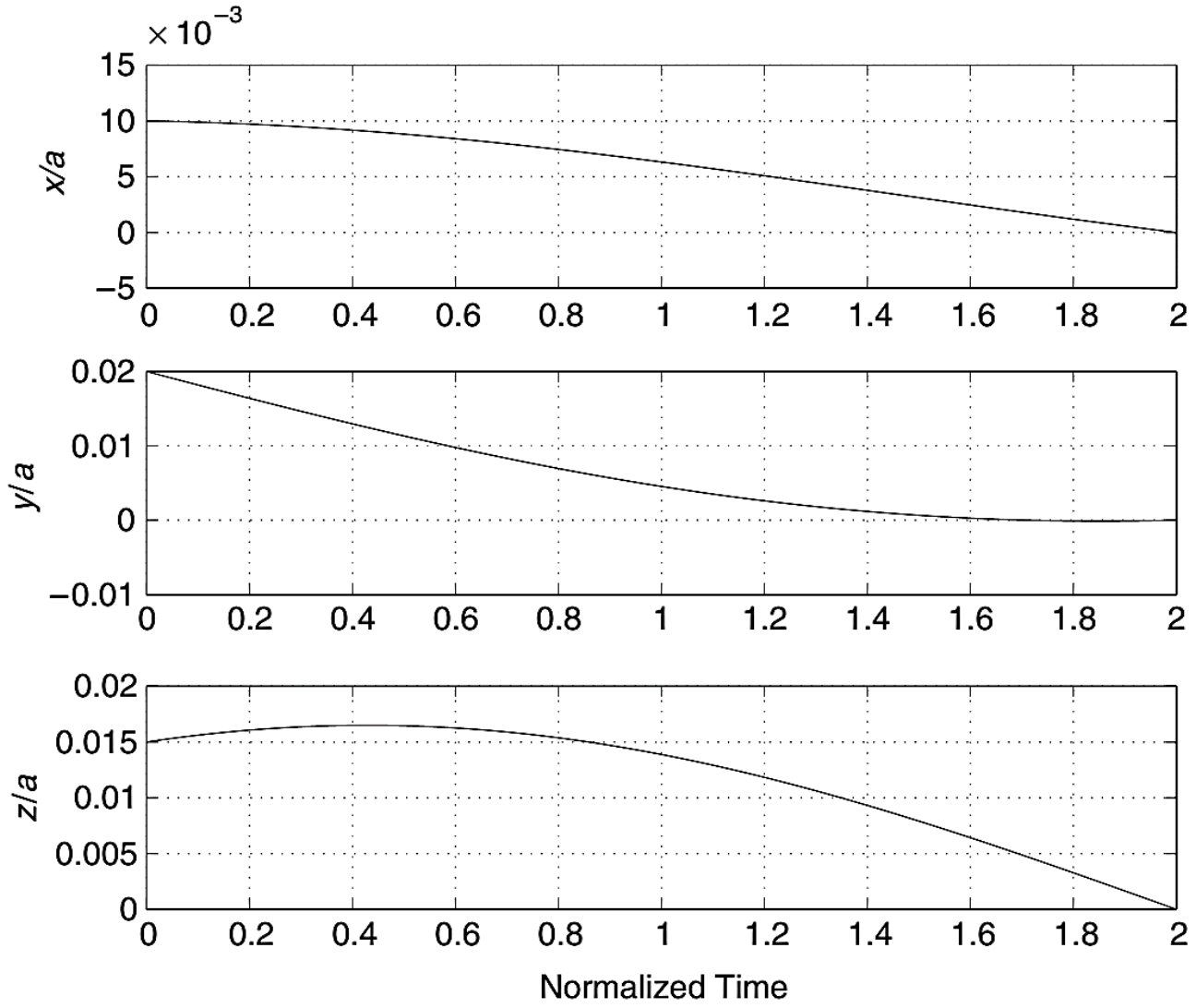


Fig. 4.5 – Normalized relative position components (Alfriend et al, 2010).

Two-Impulse Linear Rendezvous

- Figure 4.6 shows the **relative velocity components** and the **impulsive velocity correction** at **impact**, which **annuls** the final values of the relative velocity to yield **perfect rendezvous**.

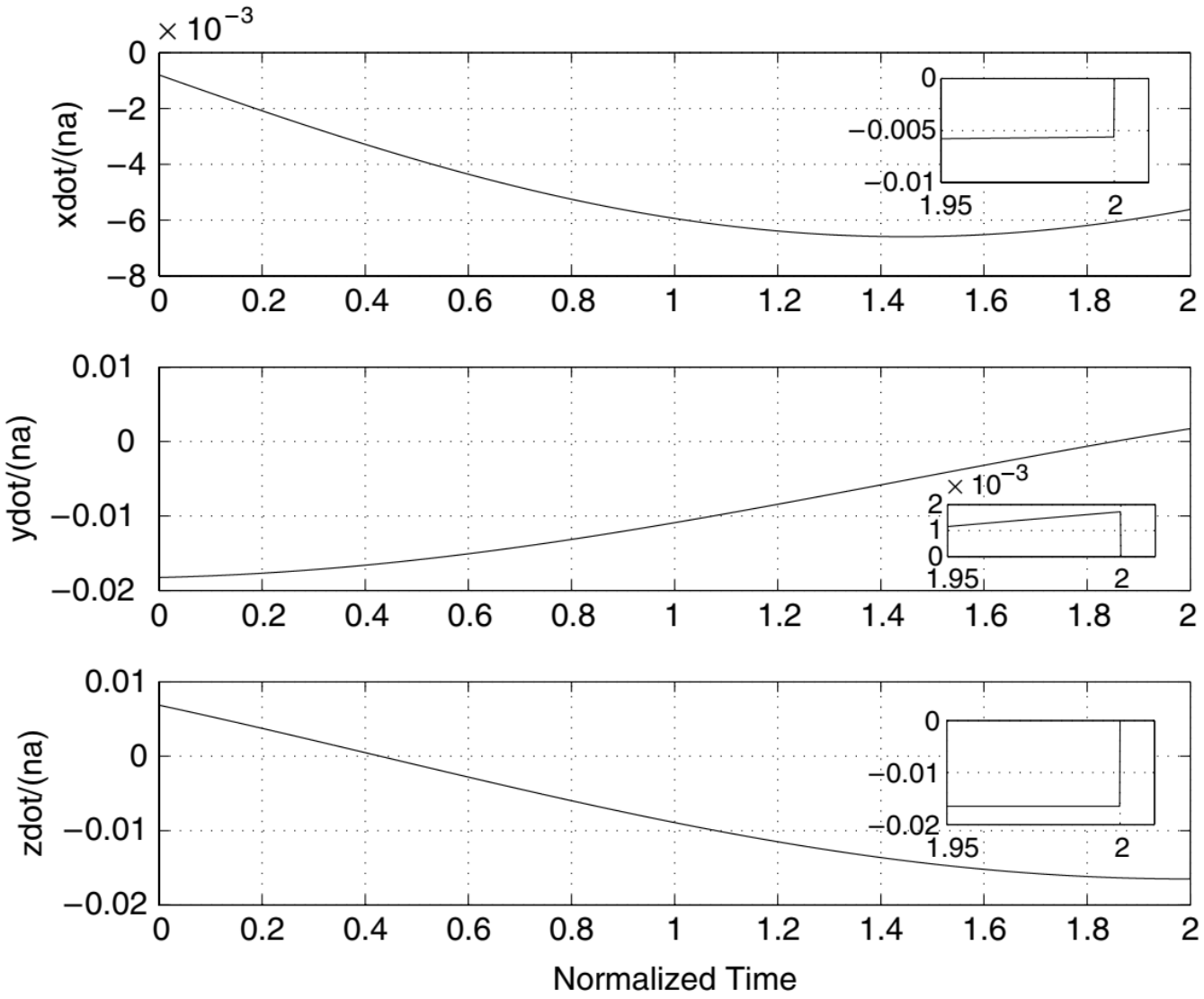


Fig. 4.6 – Normalized relative velocity components (Alfriend et al, 2010).

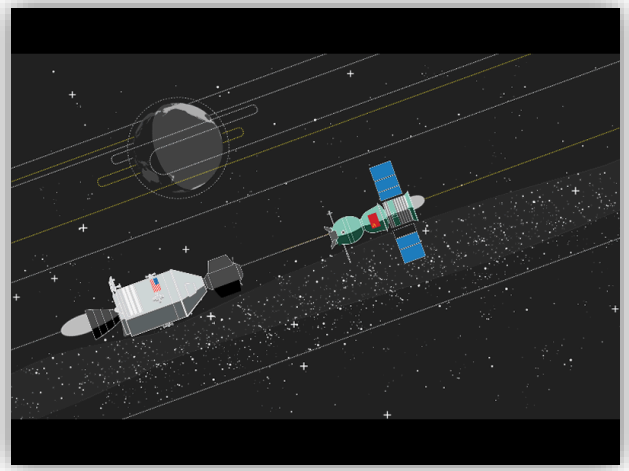
Two-Impulse Linear Rendezvous

- From the previous discussion, it is obvious that the **total velocity change** required for rendezvous **strongly depends on the time of flight**.

- We can thus ask whether there exists a t_f for which the **required** Δv is **minimal** given the **initial conditions** provided in this example.

- We may attempt to answer this query **analytically**, perform a **batch of simulations** or **simulate the adjoint system** to discover the **optimum flight time**.

- Adopting the **latter approach**, we plotted the **total** Δv vs. t_f and obtained the **desired optimum** t_f (in **normalized** units).



Two-Impulse Linear Rendezvous

- Figure 4.7 shows a **plot** of $\Delta\bar{v}$ vs. $\bar{t}_f$.

- $\Delta\bar{v}$ is **maximal near the singularities** mentioned previously.

- The **minimal velocity change**, $\Delta\bar{v} = 0.0375$, is achieved at $\bar{t}_f \approx 4.65$.

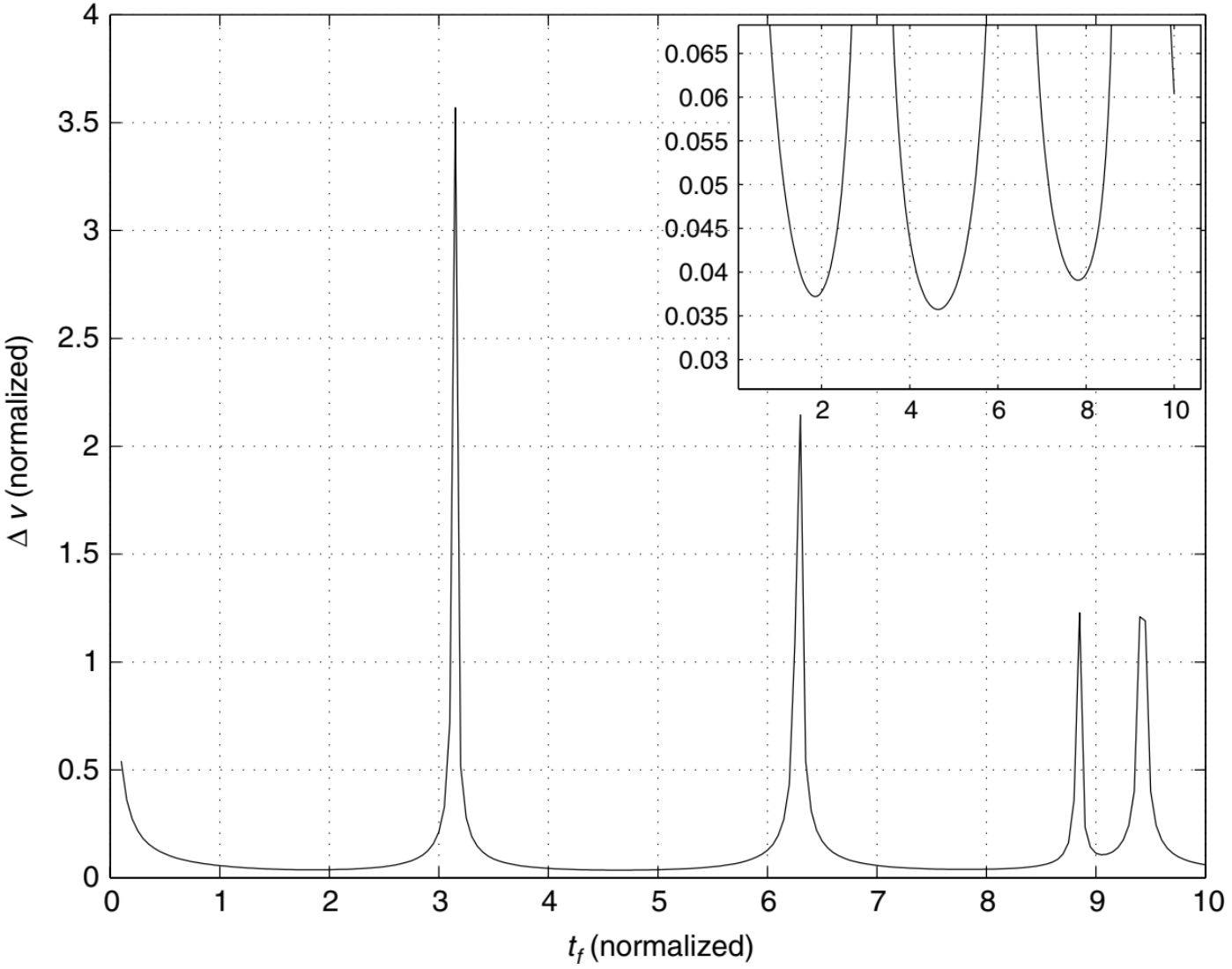


Fig. 4.7 – The total velocity change required for rendezvous varies with the time of flight (Alfriend et al, 2010).

Two-Impulse Linear Rendezvous



- Note that the **rendezvous methodology** implemented in this class relies on the **CW equations**, and is hence **limited to circular unperturbed target orbits**.

- More **general solutions for the rendezvous problems** have been obtained, see e.g. the Ref. Bond (1999) (and references therein), in which a **bounded solution for the rendezvous problem** was derived by **canceling** **second-order eccentricity terms** in the **approximation** of the chaser's orbit.

Summary Lecture #4



- Introduction
- The Clohessy–Wiltshire Equations
- Two-impulse Linear Rendezvous
- **Accommodating Second-Order Nonlinearities**
- Curvilinear vs. Cartesian Relative Coordinates
- Elliptic Reference Orbits
- Review and References



Accommodating Second-Order Nonlinearities

- Equations (3.75)–(3.77) can be **approximated** by either a **second-order expansion of the differential gravitational acceleration about a circular orbit**, or by **taking third-order expressions in the Legendre-polynomial expansion** of the Lagrangian.

$$\ddot{x} - 2n_0\dot{y} - n_0^2 x = -\frac{\mu(a_0 + x)}{\left[(a_0 + x)^2 + y^2 + z^2\right]^{\frac{3}{2}}} + \frac{\mu}{a_0^2} \qquad \text{Eq. (3.75)}$$

$$\ddot{y} + 2n_0\dot{x} - n_0^2 y = -\frac{\mu y}{\left[(a_0 + x)^2 + y^2 + z^2\right]^{\frac{3}{2}}} \qquad \text{Eq. (3.76)}$$

$$\ddot{z} = -\frac{\mu z}{\left[(a_0 + x)^2 + y^2 + z^2\right]^{\frac{3}{2}}} \qquad \text{Eq. (3.77)}$$

- This **procedure** leads to the following equations, written using **normalized coordinates and non-dimensional time**:

$$\bar{x}'' - 2\bar{y}' - 3\bar{x} = \left[\frac{\bar{y}^2}{2} + \frac{\bar{z}^2}{2} - \bar{x}^2\right] \qquad \text{Eq. (4.44)}$$

$$\bar{y}'' + 2\bar{x}' = 3\bar{x}\bar{y}$$

$$\bar{z}'' = 3\bar{x}\bar{z}$$

Accommodating Second-Order Nonlinearities

- A **perturbation method** for **accommodating** the **effects of second-order nonlinearities** in the **design of periodic relative orbits** was proposed by Vaddi et al. (2003).
- They obtained a **first-order perturbation solution** to Eq. (4.44) and showed that the **initial conditions provided by the CW solutions** can be **corrected** to achieve **bounded relative orbits**.
- Their **correction equation**, in **dimensional form**, is

$$\dot{y}(0) + 2nx(0) + \frac{3n}{2a}\left(2{\rho_x}^2 + 2{\rho_y}^2 + {\rho_z}^2 + 6\rho_x\rho_y\cos\alpha_x + 3{\rho_x}^2\cos 2\alpha_x\right) = 0 \qquad \text{Eq. (4.45)}$$

  where ρ_x, ρ_y, and ρ_z are the **constants** describing the **CW solutions**, as discussed earlier in Eqs. (4.24)–(4.26).

- Note that α_z does **not affect** the above **solution**.

Accommodating Second-Order Nonlinearities

- Equation (4.45) can be **satisfied** by **selecting** either $x(0)$ or $\dot{y}(0)$.

- Furthermore, Eq. (4.45) can also be **utilized** to **estimate** the **effect of nonlinearity on secular along-track growth**.

- If Eq. (4.23) is **satisfied**, an **estimate of the along-track drift rate due to nonlinearity**, is

Eq. (4.23)

$$\dot{y}(0) = -2nx(0)$$

$$\dot{y}_{\text{drift}} = -\frac{3n}{2a}(2\rho_x{}^2 + 2\rho_y{}^2 + \rho_z{}^2 + 6\rho_x\rho_y\cos\alpha_x + 3\rho_x{}^2\cos 2\alpha_x) \qquad \text{Eq. (4.46)}$$

- Specializing Eq. (4.46) to the case of a **PCO** with $2\rho_x = \rho_z \equiv \rho$, $\rho_y = 0$, and $\alpha_x = \alpha_z \equiv \alpha$, a **linear estimate of the along-track drift per orbit** is

$$y_{\text{drift}} = -\frac{9\pi\rho^2}{4a_0}(2 + \cos 2\alpha) \qquad \text{Eq. (4.47)}$$

Accommodating Second-Order Nonlinearities

EXAMPLE:

- For a **circular orbit** with $a_0 = 7100$ km, $\rho = 1$ km, the **drift rate** is approximately equal to 3 m per orbit, for $\alpha = 0°$ and 1 m per orbit, for $\alpha = 90°$.

- A **perturbation solution** based on Eq. (4.44) for the **differential energy between two satellites** in a formation can be **derived** from the **more general results** of Sengupta et al. (2006) as follows:

$$\delta\bar{\mathcal{E}} = (\bar{y}' + 2\bar{x}) + \frac{1}{2}\left[(\bar{x}' - \bar{y})^2 + (\bar{y}' + \bar{x})^2 + (\bar{z}')^2 - \left(2\bar{x}^2 - \bar{y}^2 - \bar{z}^2\right)\right]$$

Eq. (4.48)

- An **alternate approach**, followed by Jiang et al. (2007), is to **approximate** the **differential semimajor axis**, δa, by a **Taylor series expansion** of a_1 (calculated from Eq. (3.35)) about the **semimajor axis of the chief**.

Eq. (3.35)

$$\mathcal{E}_1 = \frac{1}{2}v_1^2 - \frac{\mu}{r_1} = \frac{1}{2}\left[v_x^2 + v_y^2 + v_z^2\right] - \frac{\mu}{r_1}$$
$$= \frac{1}{2}\{(\dot{x} - \dot{\theta}_0 y + \dot{r}_0)^2 + [\dot{y} + \dot{\theta}_0(x + r_0)]^2 + \dot{z}^2\} - \frac{\mu}{\sqrt{(r_0 + x)^2 + y^2 + z^2}}$$

Accommodating Second-Order Nonlinearities

- For example, a **linear approximation valid for circular reference orbits**, written in **non-dimensional coordinates**, is:

$$\delta\bar{a}^{(1)} = 2(\bar{y}' + 2\bar{x}) \qquad \text{Eq. (4.49)}$$

- A **second-order expansion** for δa is

$$\delta\bar{a}^{(2)} = \delta\bar{a}^{(1)} + \left[\left(\delta\bar{a}^{(1)}\right)^2 + (\bar{x}' - \bar{y})^2 + (\bar{y}' + \bar{x})^2 + \bar{z}^2 - \left(2\bar{x}^2 - \bar{y}^2 - \bar{z}^2\right)\right] \qquad \text{Eq. (4.50)}$$

- It can be shown by **expanding** the **normalized form of the energy equation** $(\varepsilon = -\mu/2a)$ that Eqs. (4.48) and (4.50) are related as

$$\boxed{\delta\bar{a}^{(2)} \approx 2\left[\delta\bar{\mathcal{E}} + 2\left(\delta\bar{\mathcal{E}}\right)^2\right]} \qquad \text{Eq. (4.51)}$$

- For **circular reference orbits**, a **linear estimate of the non-dimensional along-track drift rate due to second-order nonlinear effects** can be obtained from the **non-dimensional** δa:

$$\boxed{\bar{y}' = -\frac{3}{2}\delta\bar{a}} \qquad \text{Eq. (4.52)}$$

Summary Lecture #4



- Introduction
- The Clohessy–Wiltshire Equations
- Two-impulse Linear Rendezvous
- Accommodating Second-Order Nonlinearities
- **Curvilinear vs. Cartesian Relative Coordinates**
- Elliptic Reference Orbits
- Review and References



Curvilinear vs. Cartesian Relative Coordinates

- The **CW equations** express the **relative motion** with respect to a **rotating Cartesian coordinate system** whose **origin** coincides with the **chief satellite**.

- The **assumptions** made in their **derivation** are that



  (a) the reference of chief orbit is **circular**,

  (b) the Earth is **spherically symmetric**, and

  (c) the **distance** between the chief and follower is **small** compared to the radius of the orbit so that **only the linear terms in the expansion of the differential gravity are retained**.

- In this section, we explore some **effects of the linearization assumptions** and show **how to minimize the effect of the neglected nonlinear terms** using a **change in variables**.

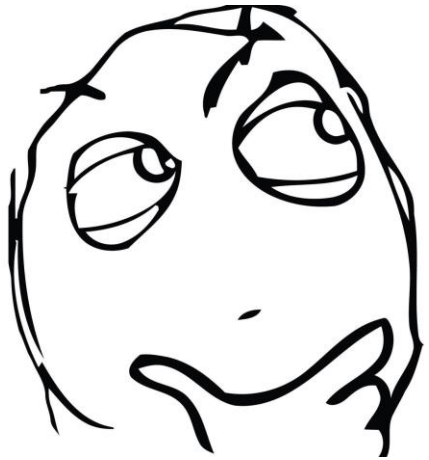
Curvilinear vs. Cartesian Relative Coordinates

- In the **derivation of the CW equations**, a **linearization** is performed in the **expansion** of the **differential gravity**.

- However, the **initial conditions are also affected**; this happens when:

    i. obtaining the **relative initial conditions** from the **exact inertial initial conditions** of both satellites, or

    ii. when obtaining the **exact initial conditions** of the follower from the **chief's initial conditions** and the **follower's relative initial conditions**.

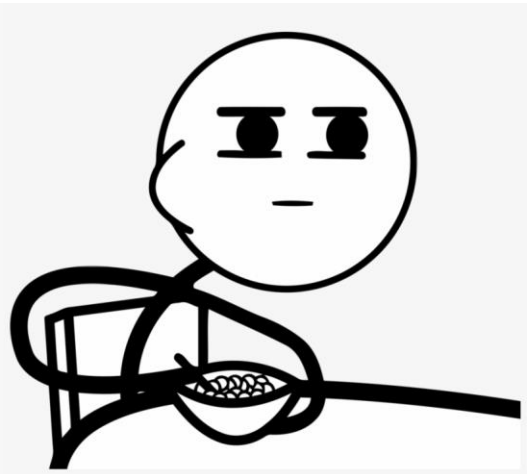


Curvilinear vs. Cartesian Relative Coordinates

- To illustrate this **initial condition impact**, consider a **simple example** of **co-orbital motion**.

- The **chief** is in a **circular orbit** of **radius** a_0 and the **follower** is in the same orbit, but **leading** the chief by an **angle** ζ, as shown in Fig. 4.8.

- Now, let's obtain the **exact relative initial conditions** for the **follower**.



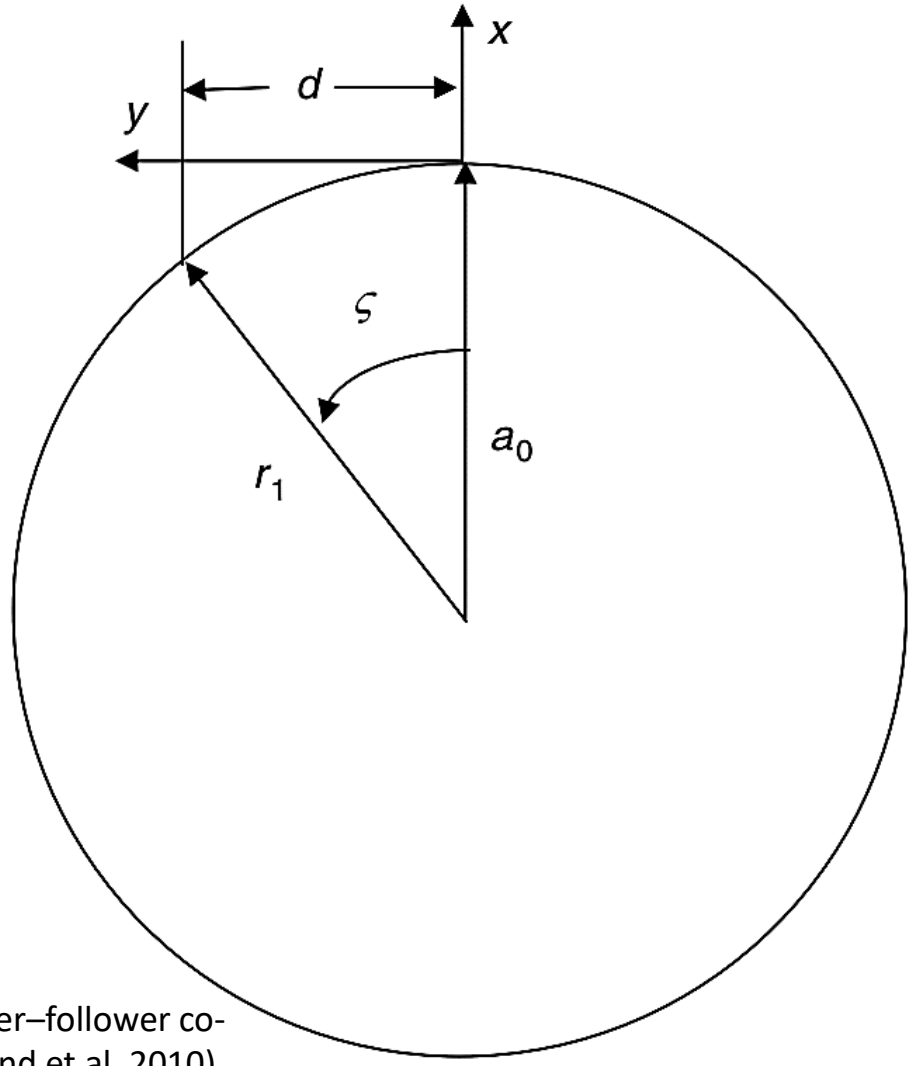


Fig. 4.8 –A simple example of leader–follower co-orbital formation (Alfriend et al, 2010).

Curvilinear vs. Cartesian Relative Coordinates

- To that end, we first establish an **inertial coordinate system**, (X, Y, Z), whose axes are **aligned** at this instant with the **LVLH system** (x, y, z), but its **origin** is at the **center of the Earth**.

- The **initial conditions of the follower** in the **inertial** $\mathcal{I}$ frame are

$$
\begin{aligned}
X(0) &= a_0 \cos\varsigma, \quad \dot{X}(0) = -a_o n \sin\varsigma \\
Y(0) &= a_0 \sin\varsigma, \quad \dot{Y}(0) = a_0 n \cos\varsigma \\
Z(0) &= 0, \quad \dot{Z}(0) = 0
\end{aligned}
$$

Eq. (4.53)

- The **corresponding initial conditions** in the **relative LVLH frame** are

$$
\begin{aligned}
x(0) &= a_0 (\cos\varsigma - 1), \quad \dot{x}(0) = 0 \\
y(0) &= a_0 \sin\varsigma, \quad \dot{y}(0) = 0 \\
z(0) &= 0, \quad \dot{z}(0) = 0
\end{aligned}
$$

Eq. (4.54)

Curvilinear vs. Cartesian Relative Coordinates

- The **differential semimajor axis** resulting from these **initial conditions** is

$$\delta a = 4x(0) + 2\dot{y}(0)/n_0 = 4a_0(\cos\varsigma - 1) \approx -2a_0\varsigma^2$$

Eq. (4.55)

- Thus, the **secular term coefficient** in Eq. (4.18) is **not zero**.

Eq. (4.18)

$$y(t) = -[6nx(0) + 3\dot{y}(0)]t + \left[y(0) - \frac{2\dot{x}(0)}{n}\right] + \left[6x(0) + \frac{4\dot{y}(0)}{n}\right]\sin(nt) + \frac{2\dot{x}(0)}{n}\cos(nt),$$

- The **secular in-track drift per orbit** is

$$3\pi\delta a = 12\pi a_0(1 - \cos\varsigma) \approx 6\pi a_0\varsigma^2$$

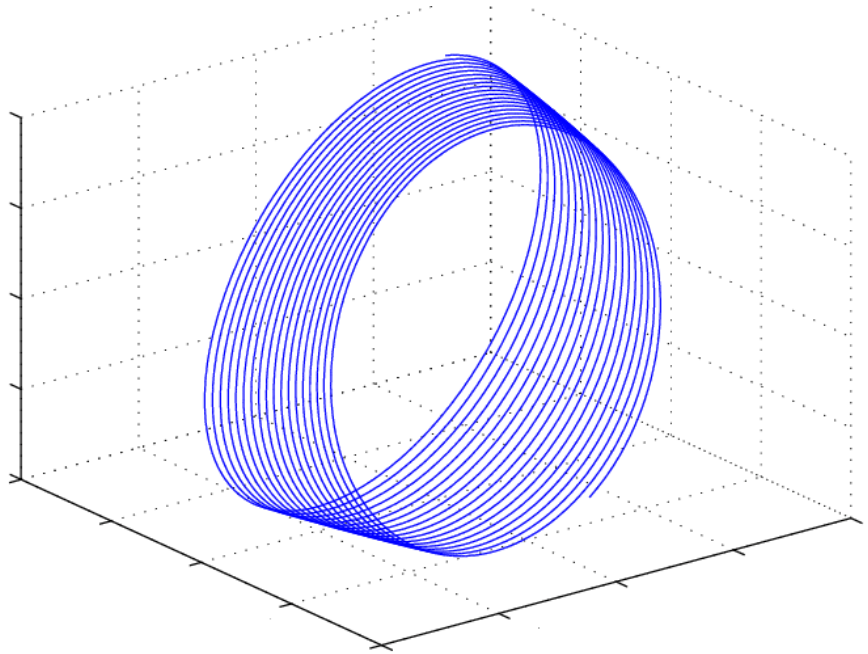
Eq. (4.56)

Curvilinear vs. Cartesian Relative Coordinates

EXAMPLE 4.3

- Evaluate the **drift per orbit** **caused** by the **CW initial conditions** for an **initial along-track separation** and a **circular chief orbit** of $a_0 = 7000$ km based on Eq. (4.57).

$$3\pi\,\delta a = 12\pi\,a_0(1-\cos\varsigma) \approx 6\pi\,a_0\varsigma^2 \qquad \text{Eq. (4.57)}$$





Curvilinear vs. Cartesian Relative Coordinates

Solution of Example 4.3

- Figure 4.9 shows the **along-track drift per orbit** that **occurs** when using the **relative Cartesian reference frame** for selecting the **initial conditions**.
- A **drift of 269 m/orbit** occurs for a **10 km initial separation**.
- This **drift** would have to be **canceled** **by the control**.

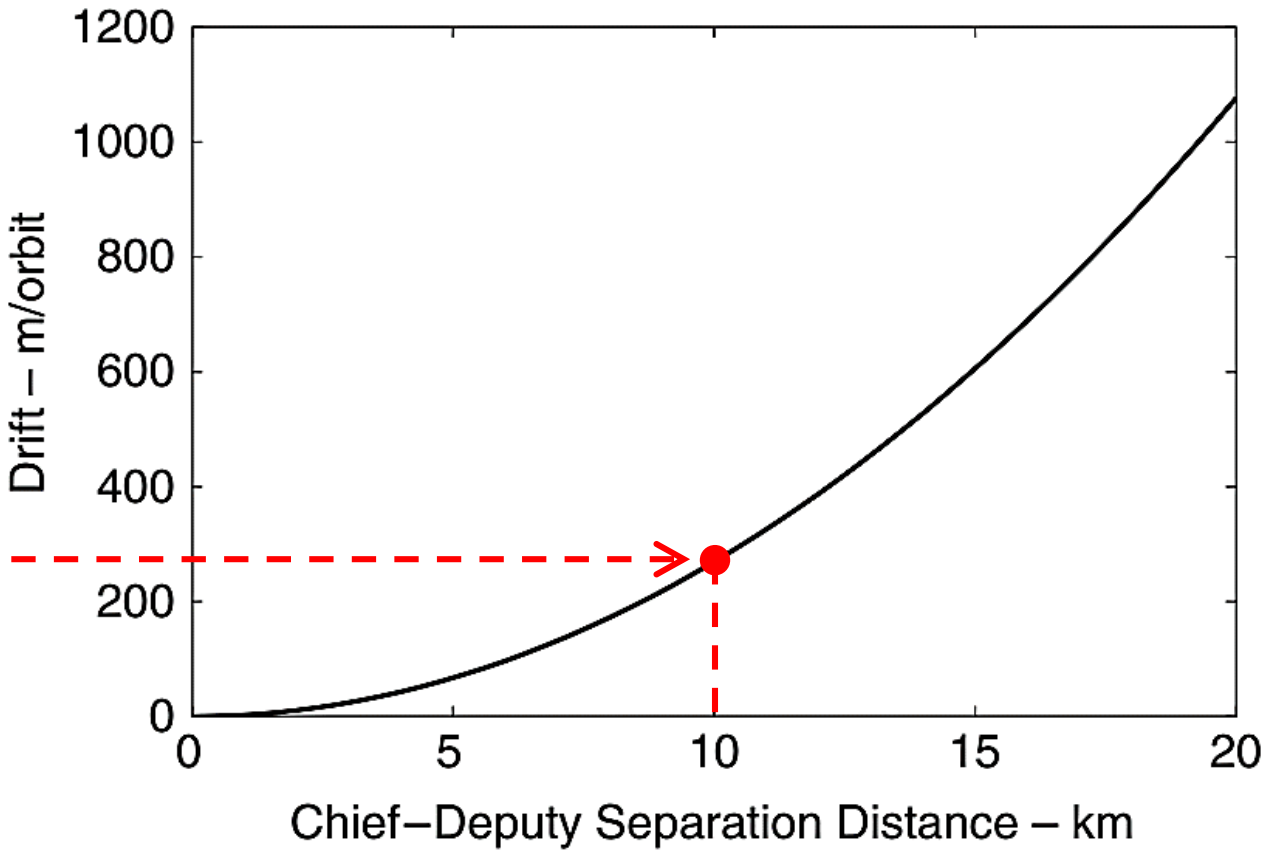


Fig. 4.9 –Drift per orbit caused by the CW equations initial conditions (Alfriend et al, 2010).

Curvilinear vs. Cartesian Relative Coordinates

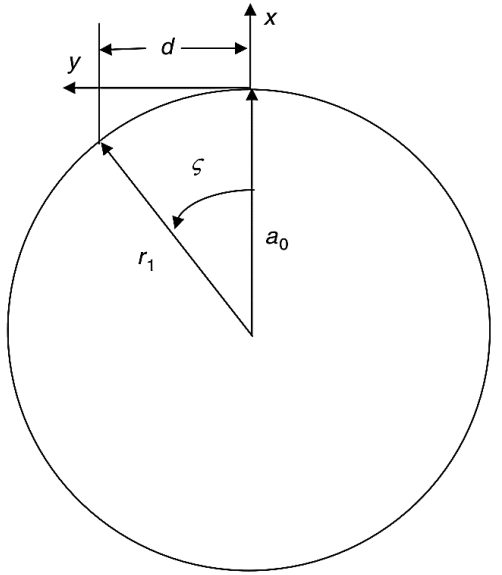
- The **inverse scenario**, that of **starting** with the **initial relative state** and **obtaining** the **initial state of the follower** in the $\mathcal{I}$ frame, has a **similar problem**.

- With the same problem of a **leader–follower formation** with an **along-track separation** $D \approx a_0\zeta$, as shown in Fig. 4.8, the **radius of the follower's orbit** is

$$r_1 = \left(a_0^2 + D^2\right)^{1/2} \approx a_0\left[1 + 0.5\,(D/a_0)^2\right] \qquad \text{Eq. (4.58)}$$

- Thus

$$\delta r \approx 0.5 a_0 \varsigma^2 \qquad \text{Eq. (4.59)}$$



- The **follower velocity** is

$$v = \left[(a_0 n_0)^2 + (D n_0)^2\right] \qquad \text{Eq. (4.60)}$$

Curvilinear vs. Cartesian Relative Coordinates

- From the **vis-viva equation**,

$$-\frac{\mu}{a} = v^2 - 2\frac{\mu}{r}$$

we can write

$$\frac{\mu}{a_0^2}\delta a = 2v\delta v + 2\frac{\mu}{r_0^2}\delta r \qquad \text{Eq. (4.61)}$$

and hence

$$\boxed{\delta a = 2a_0 \sin \varsigma^2 \approx 2a_0\varsigma^2} \qquad \text{Eq. (4.62)}$$

- Comparing to Eq. (4.55), we see that δa has the **same magnitude**, but **opposite sign**.

Eq. (4.55)

$$\boxed{\delta a = 4x(0) + 2\dot{y}(0)/n_0 = 4a_0(\cos\varsigma - 1) \approx -2a_0\varsigma^2}$$

Curvilinear vs. Cartesian Relative Coordinates

- Therefore, the same **DRIFT** occurs **whether or not** the **follower's relative initial conditions** are **obtained** from:
  i. the **exact initial conditions of the chief and follower**,
  ii. the **chief's exact and follower's relative initial conditions**.

- If the **initial radius** and the **corresponding initial relative states** are **used** as the **initial conditions** for **numerical integration or an analytic theory**, the **two satellites will slowly drift apart**, even though they are in the **same orbit**.

- Thus, the use of the **Cartesian rotating coordinate system** results in **undesirable errors** in the **solution** due to the **nonlinear terms**.

- These **nonlinear effects** can be **reduced** if a **curvilinear coordinate system** is used.

Curvilinear vs. Cartesian Relative Coordinates

- To that end, let $(\delta r, \theta_r, \phi_r)$ be the **relative coordinates**, as shown in Fig. 4.10, where:

δr is the **difference** between the **radii of the follower and chief**,

θ_r is the **angle** between the **projection of the follower's radius vector** in the chief's orbit plane and the **chief's radius vector**, and

ϕ_r is the **angle** between **follower's vector** and its projection in the **chief's orbit plane**.

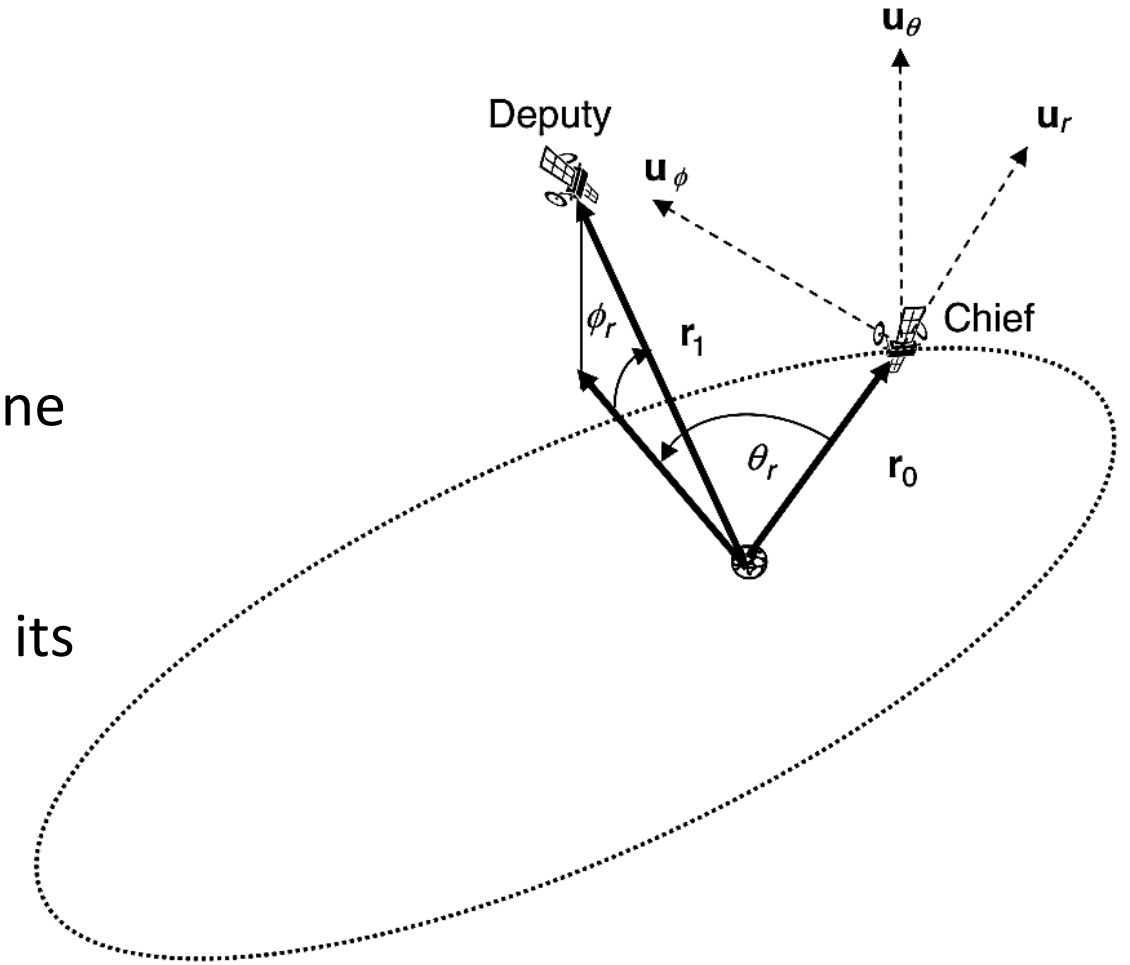


Fig. 4.10 – Relative curvilinear coordinate system (Alfriend et al, 2010).

Curvilinear vs. Cartesian Relative Coordinates

- Thus, $(\delta r, a_0\theta_r, a_0\phi_r)$ are **analogous** to (x, y, z).

- The **radius and velocity vectors of the follower** are

$$\mathbf{r}_1 = r_1\hat{\mathbf{r}}_1 = (a_0 + \delta r)\,\hat{\mathbf{r}}_1 \qquad \text{Eq. (4.63)}$$

$$\mathbf{v}_1 = \delta\dot{r}\hat{\mathbf{r}}_1 + (a_0 + \delta r)\,\boldsymbol{\omega}_1 \times \hat{\mathbf{r}}_1 \qquad \text{Eq. (4.64)}$$

where

$$\boldsymbol{\omega}_1 = \boldsymbol{\omega}_0 + \boldsymbol{\omega}_{1/0}$$

$$\boldsymbol{\omega}_0 = n_0\hat{\mathbf{u}}_\phi \qquad \text{Eq. (4.65)}$$

$$\boldsymbol{\omega}_{1/0} = \dot{\theta}_r\hat{\mathbf{u}}_\phi + \dot{\phi}_r\left(\hat{\mathbf{u}}_\theta\cos\theta_r - \hat{\mathbf{u}}_r\sin\theta_r\right)$$

$$\hat{\mathbf{r}}_1 = \hat{\mathbf{u}}_r\cos\phi_r\cos\theta_r - \hat{\mathbf{u}}_\theta\cos\phi_r\sin\theta_r - \hat{\mathbf{u}}_\phi\sin\phi_r \qquad \text{Eq. (4.66)}$$

Curvilinear vs. Cartesian Relative Coordinates

- The **acceleration of the follower** is

$$\ddot{\mathbf{r}}_1 = \delta\ddot{r}\hat{\mathbf{r}}_1 + 2\delta\dot{r}\boldsymbol{\omega}_1 \times \hat{\mathbf{r}}_1 + r_1\dot{\boldsymbol{\omega}}_1 \times \hat{\mathbf{r}}_1 + r_1\boldsymbol{\omega}_1 \times \left(\boldsymbol{\omega}_1 \times \hat{\mathbf{r}}_1\right) \qquad \text{Eq. (4.67)}$$

where

$$\begin{aligned}\dot{\boldsymbol{\omega}}_1 = {} & \ddot{\theta}_r\hat{\mathbf{u}}_r + \ddot{\phi}_r\left(\hat{\mathbf{u}}_\theta \cos\theta_r - \hat{\mathbf{u}}_r \sin\theta_r\right) \\ & - \dot{\theta}_r\dot{\phi}_r\left(\hat{\mathbf{u}}_r \cos\theta_r + \hat{\mathbf{u}}_\theta \sin\theta_r\right) + \boldsymbol{\omega}_0 \times \boldsymbol{\omega}_{1/0}\end{aligned} \qquad \text{Eq. (4.68)}$$

- Note that in the **curvilinear coordinate formulation** there are **nonlinear terms** in the **kinematic equation**, whereas in the **cartesian formulation** the **kinematic equation** is **linear**.

- Substituting Eqs. (4.65) and (4.68) into Eq. (4.67) and **linearizing** in δr, θ_r and ϕ_r gives

$$\begin{aligned}\ddot{\mathbf{r}}_1 = {} & \left(\delta\ddot{r}_1 - 2n_0a_0\dot{\theta}_r - n_0^2\delta r - n_0^2a_0\right)\hat{\mathbf{u}}_r \\ & + \left(a_0\ddot{\theta}_r + 2a_0n_0\delta\dot{r} - a_0n_0^2\theta_r\right)\hat{\mathbf{u}}_\theta - a_0\ddot{\phi}_r\hat{\mathbf{u}}_\phi\end{aligned} \qquad \text{Eq. (4.69)}$$

Curvilinear vs. Cartesian Relative Coordinates

- The **gravitational acceleration** is

$$-\frac{\mu}{r_1^2}\hat{\mathbf{r}}_1 = -\frac{\mu}{(a_0+\delta r)^2}\hat{\mathbf{r}}_1$$

Eq. (4.70)

- **Expanding** the **denominator** in a **Taylor series** in δr, **substituting** for $\hat{\mathbf{r}}_1$ and **linearizing** in δr, θ_r and ϕ_r gives

$$-\frac{\mu}{r_1^2}\hat{\mathbf{r}}_1 = -n_0^2\left(a_0\hat{\mathbf{u}}_r - 2\delta r\hat{\mathbf{u}}_r - a_0\theta_r\hat{\mathbf{u}}_\theta - a_0\phi_r\hat{\mathbf{u}}_\phi\right)$$

Eq. (4.71)

- Equating Eqs. (4.69) and (4.71) gives the **equations of relative motion in curvilinear coordinates**:

$$\begin{aligned}
&\delta\ddot{r} - 2a_0n_0\dot{\theta}_r - 3n_0^2\delta r = 0\\
&a_0\ddot{\theta}_r - 2n_0\delta\dot{r} = 0\\
&\ddot{\phi}_r + n_0^2\phi_r = 0
\end{aligned}$$

Eq. (4.72)

- Note that the Eqs. (4.72) are **identical** to the **CW equations** with (x, y, z) being **replaced** by $(\delta r, a_0\theta_r, a_0\phi_r)$.

Curvilinear vs. Cartesian Relative Coordinates

- The **difference** is that the **along-track and cross-track distances** are **measured** along **arc lengths**; they are **not linear displacements**.

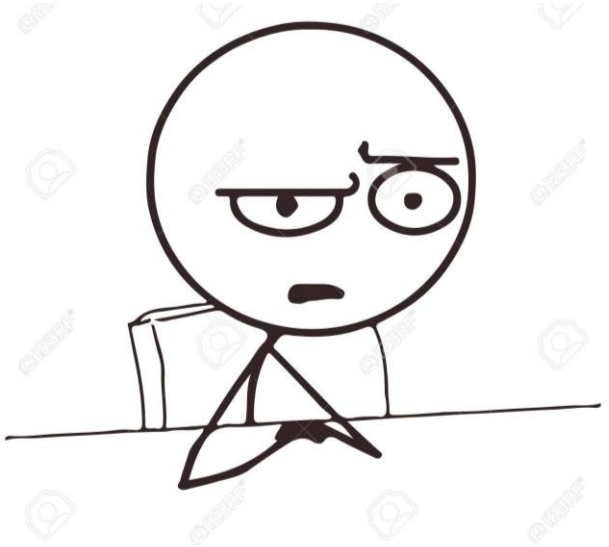
- This **reduces the nonlinear effects** in the **computation** of the **initial conditions**.

- **Another way to look** at this (when using the **CW equations**) is to let:

  x be the **radii difference**,

  y be the **arc length displacement** along the **chief orbit**, and

  z be the **arc length displacement** in the **cross-track direction**.



Curvilinear vs. Cartesian Relative Coordinates



- Due to the **linearization** there will still be **errors** that occur with the use of the **curvilinear coordinate equations**, but these **errors are much smaller**.

- In addition to the **effect of the neglected nonlinearities** in the **initial conditions**, there are the **neglected nonlinearities** that occur from the **expansion** of the **differential gravity**.

- The **use** of the **curvilinear coordinates** also **reduces** these **effects**.

Summary Lecture #4



- Introduction
- The Clohessy–Wiltshire Equations
- Two-impulse Linear Rendezvous
- Accommodating Second-Order Nonlinearities
- Curvilinear vs. Cartesian Relative Coordinates
- **Elliptic Reference Orbits**
- Review and References



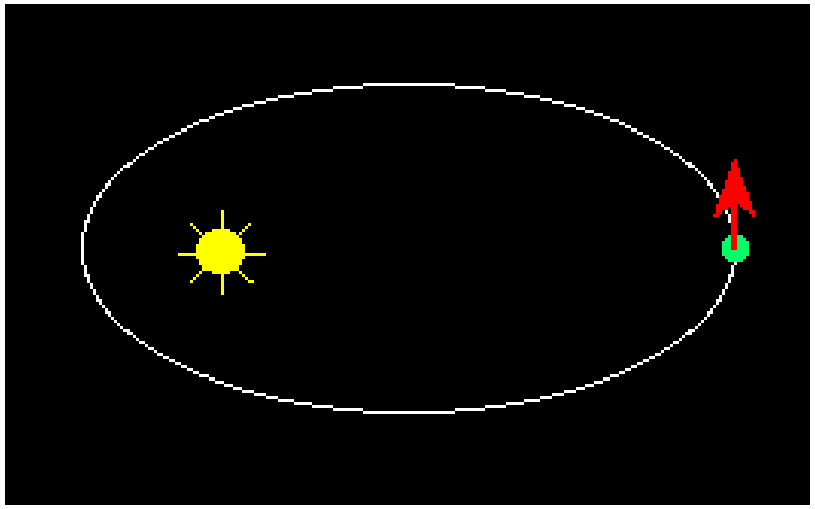
Elliptic Reference Orbits

- For **circular reference orbits**, the choice of the **independent variable** of either **time** or **true anomaly** leads to **the same form** of the **equations of motion**.

- This is **not** the case for **elliptic reference orbits**.

- **Time-explicit solutions** are **attractive** since **Kepler's equation** need **not be solved** to produce **results** at **specified points of time**.

- However, **true-anomaly-based solutions** have certain **advantages**, especially **near the perigee**, and are more **suitable** for **long-term motion prediction** for **elliptic orbits**.



Elliptic Reference Orbits

- Alfriend et al. (2010) presented several **methods for the propagation of relative motion in the elliptic orbit case**:
  - **Melton's STM (State Transition Matrix)**
  - **Lawden and Tschauner-Hempel Equations**
  - Carter's STM
  - Yamanaka and Ankersen's STM
  - Broucke's STM
  - STM of Lee, Cochran, and Jo
  - Nazarenko's STM



Elliptic Reference Orbits

Review of State Transition Matrix (STM)

- In **control theory**, the **state-transition matrix** is a **matrix** whose **product** with the **state vector** x at an **initial time** t_0 **gives** x at a **later time** t.

- The **state-transition matrix** can be used to obtain the **general solution of linear dynamical systems**.

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)\mathbf{u}(t),\ \mathbf{x}(t_0) = \mathbf{x}_0$$

where $\mathbf{x}(t)$ are the **states of the system**, $\mathbf{u}(t)$ is the **input signal**, and $\mathbf{x}_0$ is the **initial condition** at t_0.

- Using the **state-transition matrix** $\mathbf{\Phi}(t,\tau)$, the **solution** is given by

$$\mathbf{x}(t) = \mathbf{\Phi}(t,t_0)\mathbf{x}(t_0) + \int_{t_0}^{t} \mathbf{\Phi}(t,\tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau$$

Elliptic Reference Orbits

Time as the independente variable: Melton's STM

- Melton (2000) has developed **expansions** of the **STM** in **powers of eccentricity**, in both **Cartesian** as well as **cylindrical coordinate systems**.

- The **linearized equations of relative motion with respect to an unperturbed eccentric reference orbit** (chief) in a **Cartesian coordinate system** can be obtained from Eqs. (3.14)–(3.16) as follows

$$\ddot{x} - 2\dot{\theta}\dot{y} - \ddot{\theta}y - \dot{\theta}^2 x = \frac{2\mu x}{r^3} \qquad \text{Eq. (4.73a)}$$

$$\ddot{y} + 2\dot{\theta}\dot{x} + \ddot{\theta}x - \dot{\theta}^2 y = -\frac{\mu y}{r^3} \qquad \text{Eq. (4.73b)}$$

$$\ddot{z} = -\frac{\mu z}{r^3} \qquad \text{Eq. (4.73c)}$$

Elliptic Reference Orbits

- For **unperturbed motion**, using the **result**

$$r^2 = \frac{h}{\dot{\theta}} \qquad \text{Eq. (4.74)}$$

where h is the **angular momentum**, Eqs. (4.73) can also be **written** as

$$\ddot{x} = \left[\dot{\theta}^2 + \frac{2\mu}{h^{3/2}}\dot{\theta}^{3/2}\right] x + \ddot{\theta} y + 2\dot{\theta}\dot{y} \qquad \text{Eq. (4.75a)}$$

$$\ddot{y} = -\ddot{\theta} x + \left[\dot{\theta}^2 - \frac{\mu}{h^{3/2}}\dot{\theta}^{3/2}\right] y - 2\dot{\theta}\dot{x} \qquad \text{Eq. (4.75b)}$$

$$\ddot{z} = -\left[\frac{\mu}{h^{3/2}}\dot{\theta}^{3/2}\right] z \qquad \text{Eq. (4.75c)}$$

Elliptic Reference Orbits

- The **equations of motion** in the **cylindrical coordinate system** are **obtained** from Eqs. (4.75) by using **non-dimensional in-plane variables** $\bar{x} = x/r$ and $\bar{y} = y/r$.

- Regardless of the coordinate system utilized, the **linearized equations** can be written in **matrix form** as follows:

$$\dot{\mathbf{x}}(t) = A(t)\mathbf{x}(t) \qquad \text{Eq. (4.76)}$$

where the **state vector** is **defined** as

$$\mathbf{x} = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^T \qquad \text{Eq. (4.77)}$$

Elliptic Reference Orbits

- The **variables** r and $\dot{\theta}$ can be **expanded** in **powers of eccentricity**, as shown in Battin (1999):

$$r = a[1 - e\cos M - 1/2e^2(\cos 2M - 1) + \cdots] + \mathcal{O}(e^3) \qquad \text{Eq. (4.78a)}$$

$$\dot{\theta} = n[1 + 2e\cos M + 5/2e^2\cos 2M] + \mathcal{O}(e^3) \qquad \text{Eq. (4.78b)}$$

- Since $A(t)$ consists of **periodic terms**, it can also be **expanded** in **powers of** e:

$$A(t) = A_0 + eA_1(t) + e^2A_2(t) + \cdots \qquad \text{Eq. (4.79)}$$

where A_0 is a **constant matrix** and $A_{1,2,\ldots}$ are **functions of time**.

Elliptic Reference Orbits

- In a similar fashion, the **STM** can also be **expanded** as shown below:

$$\Phi(t, t_0) = \Phi_0(t, t_0) + e\Phi_1(t, t_0) + e^2\Phi_2(t, t_0) + \cdots \qquad \text{Eq. (4.80)}$$

where

$$\Phi_0(t, t_0) = \exp[A_0(t - t_0)] \qquad \text{Eq. (4.81)}$$

- Since the **STM** of Eq. (4.76) also **satisfies** the same **differential equation**, **direct substitution** of Eqs. (4.79) and (4.80) yields

$$\begin{aligned} \dot{\Phi}(t, t_0) &= A\Phi(t, t_0) \\ &= [A_0 + eA_1(t) + \cdots] \times [\Phi_0(t, t_0) + e\Phi_1(t, t_0) + \cdots] \end{aligned} \qquad \text{Eq. (4.82)}$$

Elliptic Reference Orbits

- Collecting the **coefficients of like powers of** e in Eq. (4.82) **results** in

$$\dot{\Phi}_1(t, t_0) = A_0\Phi_1(t, t_0) + A_1(t)\Phi_0(t, t_0)$$

$$\dot{\Phi}_2(t, t_0) = A_0\Phi_2(t, t_0) + A_1(t)\Phi_1(t, t_0) + A_2(t)\Phi_0(t, t_0)$$

$$\cdot$$

$$\cdot$$

$$\cdot$$

$$\dot{\Phi}_n(t, t_0) = A_0\Phi_n(t, t_0) + A_1(t)\Phi_{n-1}(t, t_0) + \cdots + A_{n-1}(t)\Phi_1(t, t_0) + A_n(t)\Phi_0(t, t_0)$$

Eq. (4.83)

Elliptic Reference Orbits

- Melton (2000) presents **expressions** for the **elements** of Φ_1 and Φ_2 **based** on the following **integrals**:

$$\Phi_1(t, t_0) = \int_{t_0}^{t} \Phi_0(t, s) A_1(s) \Phi_0(s, t_0) \mathrm{d}s$$

$$\Phi_2(t, t_0) = \int_{t_0}^{t} \Phi_0(t, s) \left[A_1(s) \Phi_1(s, t_0) + A_2(s) \Phi_0(s, t_0)\right] \mathrm{d}s$$

Eq. (4.84)

.

.

$$\Phi_n(t, t_0) = \int_{t_0}^{t} \Phi_0(t, s) \left[A_1(s) \Phi_{n-1}(s, t_0) + \cdots A_n(s, t_0) \Phi_0(s, t_0)\right] \mathrm{d}s$$

Elliptic Reference Orbits

- **Melton's solution** shows the **presence** of **secular terms** for the **in-plane motion variables**.

- **Secular terms** appear in the **along-track solution** since the **periods** of the **two satellites** are, in general, **not the same**.

- However, the **appearance** of **secular terms** in the **radial direction** is an **artifact** of the **series approximation** utilized.

- Melton (2000) presents **numerical results for eccentricities** in the **range** $0 \leq e \leq 0.3$.

- These **results** clearly show the **successive improvements achieved** due to the **first** and **second-order terms** in the **STM**.

Elliptic Reference Orbits

Lawden and Tschauner–Hempel Equations

- **Relative motion equations with true anomaly** as the **independent variable** were derived in Eqs. (3.28)–(3.30).

- Lawden (1963), during his studies on **optimal orbit transfers**, developed a set of equations for the so-called **primer vector**, which can be used to describe **perturbed motion with respect to elliptic orbits.**

$$\bar{x}'' - 2\bar{y}' = \frac{\partial \mathcal{W}}{\partial \bar{x}} \qquad \text{Eq. (3.28)}$$

$$\bar{y}'' + 2\bar{x}' = \frac{\partial \mathcal{W}}{\partial \bar{y}} \qquad \text{Eq. (3.29)}$$

$$\bar{z}'' = \frac{\partial \mathcal{W}}{\partial \bar{z}} \qquad \text{Eq. (3.30)}$$

- Later, DeVries (1963) and Tschauner and Hempel (TH) (1964), independently arrived at the very same equations, while working respectively, on the **problems of relative motion of particles and terminal rendezvous of spacecraft**.

Elliptic Reference Orbits

- The **equations** presented here are referred to as the **Tschauner–Hempel equations**:

$$\bar{x}'' = \frac{3}{k}\bar{x} + 2\bar{y}' \qquad \text{Eq. (4.85a)}$$

$$\bar{y}'' = -2\bar{x}' \qquad \text{Eq. (4.85b)}$$

$$\bar{z}'' = -\bar{z}' \qquad \text{Eq. (4.85c)}$$

where $(\cdot)'$ and $(\cdot)''$ indicate, respectively, the **first** and **second derivatives** with respect to f, the **true anomaly** and $k = 1 + e\cos f$.

- As can be seen from the above equations, the **in-plane and out-of-plane equations** are **decoupled** from **each other**.

Elliptic Reference Orbits

- Equation (4.85b) can be **integrated** once to **produce** $\bar{y}'$, which, when **substituted** into Eq. (4.85a), gives

$$\bar{x}'' + \left[4 - \frac{3}{k}\right]\bar{x} = 2d \qquad \text{Eq. (4.86)}$$

where d is a **constant of integration**.

- The **solution** to $\bar{x}$ **enables** one to **obtain** $\bar{y}$.

- Since Eq. (4.85c) describes a **simple harmonic motion**, its **solution** can be **easily obtained**.

- The **solution** to Eq. (4.86) was developed by Lawden (1963) and was subsequently **modified** by Carter (1998) and Yamanaka and Ankersen (2002).

Summary Lecture #4



- Introduction
- The Clohessy–Wiltshire Equations
- Two-impulse Linear Rendezvous
- Accommodating Second-Order Nonlinearities
- Curvilinear vs. Cartesian Relative Coordinates
- Elliptic Reference Orbits
- **Review and References**



Review and References

What Did We See Today?



- **Linear relative motion** models for **circular** and **elliptic** **reference orbits**.
- The famous **CW equations** have been **widely used** for **rendezvous** and **formation design and analysis**.
- The **advantages** of writing the **equations of motion in a curvilinear coordinate system** for **capturing** some of the **nonlinear effects**.
- The **treatment of the elliptic orbit case** lead us to the **Tschauner–Hempel equations** and the various forms of the **two-body STMs** for the **propagation of relative motion**.

Review and References

- BATTIN, R. H., An Introduction to the Mathematics and Methods of Astrodynamics, AIAA, Reston, VA, 1999.
- BOND, V. R., "A New Solution for the Rendezvous Problem", Advances in the Astronautical Sciences, Vol. 102, 1999, pp. 1115–1144.
- CARTER, T. E., "State Transition Matrices for Terminal Rendezvous Studies: Brief Survey and New Examples", Journal of Guidance, Control, and Dynamics, Vol. 21, No. 1, January–February 1998, pp. 148–155.
- DE VRIES, J. P., "Elliptic Elements in Terms of Small Increments of Position and Velocity Components", AIAA Journal, Vol. 1, No. 11, November 1963, pp. 2626–2629.
- JIANG, F., et al., "Approximate Analysis for Relative Motion of Satellite Formation Flying in Elliptical Orbits", Celestial Mechanics and Dynamical Astronomy, Vol. 98, No. 1, 2007, pp. 31–66.
- LAWDEN, D. F., Optimal Trajectories for Space Navigation, Butterworths, London, 1963.
- MELTON, R. G., "Time Explicit Representation of Relative Motion Between Elliptical Orbits", Journal of Guidance, Control, and Dynamics, Vol. 23, No. 4, July–August 2000, pp. 604–610.
- SENGUPTA, P., et al., "Periodic Relative Motion Near a Keplerian Elliptic Orbit with Nonlinear Differential Gravity", Journal of Guidance, Control, and Dynamics, Vol. 29, No. 5, 2006, pp. 1110-1121.
- TSCHAUNER, J. and HEMPEL, P., "Optimale Beschleunigeungsprogramme fur das Rendezvous-Manover", Astronautica Acta, Vol. 10, 1964, pp. 296–307.
- VADDI, S. S., et al., "Formation Flying: Accommodating Nonlinearity and Eccentricity Perturbations", Journal of Guidance, Control, and Dynamics, Vol. 26, No. 2, 2003, pp. 214-223.
- YAMANAKA, K. and ANKERSEN, F., "New State Transition Matrix for Relative Motion on an Arbitrary Elliptical Orbit", Journal of Guidance, Control, and Dynamics, Vol. 25, No. 1, January–February 2002, pp. 60–66.

Review and References

What will we see in the NEXT CLASS?

- How to use the **orbital elements** as constants of the unperturbed motion **instead** of the **Cartesian initial conditions**?

- What are the **advantages** of the **orbital-elements approach**?

- How to obtain **time-series approximations** of the **relative dynamics equations**?

- How to **establish** proper **initial conditions** for a **PCO-like formation**?



Review and References



Prof. Dr. ***WILLER*** *Gomes dos Santos*
Aerospace Systems Department
willer@ita.br

End

